What's Driving High Electric Vehicle Fast Charging Prices?

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Abstract

Electric vehicle fast charging is more costly than home charging, which could slow electric vehicle adoption and place higher costs on lower income households that cannot charge at home. I compile novel data on station quantities, prices, and marginal operating costs to shed light on price setting behavior across charging networks. Using a difference-in-differences approach, I show that contrary to standard theory, stations do not lower prices in response to nearby competitor entry. I estimate a model of heterogeneous consumer demand for fast charging and use the elasticities from the demand model to calculate the separate sets of social surplus maximizing prices. I show that charging prices include a large markup above what is necessary to recover fixed costs, and lowering prices to the efficient prices would increase social surplus by 7-10%. Contrary to most settings, the optimal linear and two-part tariff prices that recover fixed costs are nearly the same. In the presence of charging station capacity constraints, equating quantity demanded and supplied requires increasing efficient prices above the marginal operating costs. These results suggest that increasing competition can only reduce markups by half, and policy would need to address charging supply or inelastic consumer demand to further reduce markups.

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1 Introduction

Many of the worlds' largest car markets have set ambitious goals for electric vehicle (EV) adoption.[1] A hurdle in meeting these goals will be making EVs appealing for drivers who cannot charge at home, a population that has been resistant to purchasing EVs. Many governments have focused on public fast chargers as the solution to this challenge. For example, California estimated in 2024 that it would need 83,000 fast chargers to support a fully electric car fleet – a notable increase from the 10,000 it had in 2023 – and has earmarked around $350 million for fast charger subsidies since 2024 (Davis et al., 2024). Fast chargers can refuel an EV in 20-60 minutes (compared to 4-50 hours at other chargers), allowing drivers without access to a charger at home or work to refuel.

The premium speed comes at a premium price. Fast charging is on average three times as expensive as home charging in the United Sates (Paris, 2025). High charging prices have the potential to negatively impact both efficiency and equity. Prices above the social marginal cost could lead to deadweight loss from inefficiently low EV adoption and use for drivers without access to chargers at their home or workplace. The ability to charge at home is largely determined by whether a household has its own garage, which is highly correlated with income. In California, under a quarter of drivers who live in multi-family homes in disadvantaged or low-income communities could charge at home (Hoang and Davis, 2026).

In this paper, I explore the causes of high prices and estimate the impacts on social surplus. One barrier to research on pricing in this market is that charging networks typically share prices only through their own mobile applications, leading to an absence of centralized pricing data. I construct a novel weekly panel of hourly prices at the three largest networks in California – Tesla, EVgo, and Electrify America – by manually collecting data from networks' mobile applications. The panel includes 19 months of week by hour price data for 58% of all charging stations in California. I build a second panel of EV charging volumes and marginal operating costs in Northern California using four years of electricity consumption and rate data from Pacific Gas & Electric Company, one of the largest utilities in the United States. Combining the two datasets, I show that fast charging prices reflect large markups over marginal operating costs, which are an order of magnitude greater than those in the retail gasoline industry.

Markups are a feature of emerging markets and differentiated competition. In standard models of price competition, markups incentivize firms to enter, leading to greater competition and driving down markups. Differentiation between firms, strong consumer preferences,

[1] Although the United States has not set any targets, a number of states have their own and thirteen formed the Affordable Clean Cars Coalition to help meet their targets.

and fixed costs can all sustain markups even in the long run. I show one additional reason for markups above marginal operating costs in the fast charging market: marginal costs do not always equal marginal operating costs. Stations cannot add chargers in the short run, so the constrained marginal cost curve becomes vertical at the station's capacity. Setting prices based on the intersection of the marginal revenue and constrained marginal cost curves clears the market and ensures consumers with a high willingness to pay are not left unserved. It also results in markups above marginal operating costs. I assess the share of observed markups rationalized by each potential explanation.

I start by focusing on the effects of station openings. Contrary to results from other markets, I find that charging volumes and prices at existing stations remain unchanged following competitor entry. However, entry does decrease charging volumes by 14% at nearby stations in the same network, suggesting that consumers are willing to substitute within but not across networks. Stations do not compete on prices, and have no incentive to do so because consumers do not substitute across networks. These results indicate that subsidizing entry and increasing the number of competitors in the market is unlikely to drive down prices.

I then assess the consumer preferences, market clearing, and fixed costs explanations for markups. I estimate a system of heterogeneous consumer demand for fast charging and find very low cross-station and cross-hour price elasticities, suggesting strong consumer preferences. I use the estimated own- and cross-price elasticities to solve for social surplus and profit maximization prices. I show that all three networks set prices near the linear prices that maximize profit in the charging market, suggesting each acts as a monopolist and does not follow a two-part tariff strategy. Their behavior leads to a quantity-weighted average markup above marginal operating cost of 109%.

I calculate that equating quantity demanded and supplied to maximize social surplus requires 48% markups above marginal operating costs. Incorporating that networks need to recover fixed costs to justify entry necessitates markups of 53-98% when firms use linear prices. I show that if observed prices were lowered to the prices that would most efficiently recover fixed costs, social surplus would increase by 7-10%. My results suggest policy that lowers prices by increasing competition could increase social surplus by a large amount, though reduced profits could deter station entry. However, nearly half of the observed markups are efficient given current consumer preferences and the current supply of chargers. Policy aiming to bring fast charging prices down should both increase competition and either increase the supply of charging or shift demand to less congested times of day.

I contribute to three literatures. First, I contribute to the literature on EV charging. A number of papers have estimated the effect of charger supply on EV purchases (Remmy, forthcoming; Cole et al., 2023; Springel, 2021; Sommer and Vance, 2021; Zhou and Li, 2018;

Li et al., 2017). My results are the first on the effect of charger supply on prices and charging behavior. Importantly, I show that consumers exhibit strong network preferences that are not explained by charging standard differences. This is particularly relevant for work showing potentially large benefits of standardizing plug types, which assumes potential drivers (rather than the current EV drivers I study) choose cars based on existing stations' charging standard rather than network (Li, 2023). I also provide the first model-based estimates of demand elasticities for fast charging stations in the United States. This adds to randomized control trial evidence from Bernard et al. (2025) on elasticities of demand for public charging in the United Kingdom and Bailey et al. (2025) and Garg et al. (2026) on how drivers respond to price incentives for home and workplace charging, respectively.

Second, I contribute to the literature on market entry and competition. My results contrast the retail gasoline literature, where Taylor and Muehlegger (2025) and Davis, McRae and Seira (2025) have documented price decreases from incumbent stations in response to competitor entry. The lack of price response is rationalized by the finding that station entry does not impact charging volumes at nearby competitors. Charging volumes do decrease at nearby incumbents in the same network, indicating consumers substitute within but not across networks. These results suggests that consumer preferences or demand-side frictions, rather than the supply-side frictions Arcidiacono et al. (2020) find at grocery stores, may explain the lack of price response in the fast charging market.

Third, I contribute to the literature on fixed-cost recovery and efficient pricing. "Ramsey prices" are the set of linear prices that minimize deadweight loss while recovering fixed costs (Boiteux, 1956; Ramsey, 1927). Though they are the most efficient of all linear prices, Ramsey prices still lead to deadweight loss, and are a second best option to two-part tariffs (Feldstein, 1972; Littlechild, 1975). I modify the standard Ramsey problem to account for short-run capacity constraints and show that Ramsey prices can result in minimal deadweight loss and be nearly as efficient as two-part tariffs when capacity constraints bind.

The paper proceeds as follows. Section 2 provides information on California's fast charging market, the setting for this paper. In Section 3, I describe the data collection process and provide descriptive statistics on the fast charging market. Section 4 provides theoretical justifications for markups. In Section 5, I present estimates of the impact of station entry on charging quantities, price, and quality. I build and estimate a model of heterogeneous demand in Section 6. In Section 7, I use the elasticities from demand model to calculate profit and social surplus maximizing prices. Section 8 concludes.

2 Fast Charging

Electric vehicle charging is generally classified into three levels based on speed. Level 1 and level 2 charging take 4-10 and 40-50 hours to charge a vehicle, respectively. In this paper, I focus on level 3 charging, often called DC fast charging or simply fast charging, which can refuel a car in 20 minutes.[2] Fast charging is generally considered the infrastructure necessary to enable longer trips and provide regular refueling to those who cannot charge at home or work.

The downside to fast charging is the cost. Chargers are expensive and often require upgrades to electricity grid infrastructure before they can be connected. Stations generally cost $40,000 to $400,000 per charger depending on the speed of the charger, expense of upgrading local grid infrastructure, and number of chargers at the station (Gamage, Tal and Jenn, 2023; Li, 2023; Miyatra, 2025). An additional barrier to installation is the time it takes to upgrade the grid before chargers can be connected, which can take several months (Gamage, Tal and Jenn, 2023).

2.1 The Fast Charging Market

The fast charging market in California is highly concentrated. Stations are owned by centrally-organized charging networks that operate statewide.[3] Three networks, Tesla, EVgo, and Electrify America, operate 64% of stations.[4] Tesla, the largest of the three, builds larger stations than nearly all other networks; it operates 30% of stations and 57% of fast chargers in the state. Networks have raised consumer search costs through only listing prices on their own mobile applications. Tesla and Electrify America all share charger availability data with Google Maps, but none of the three shares prices.[5]

Another key way that stations have segmented the market is through differences in charging standards. DC fast chargers have been developed with three different charging standards.[6] All Teslas and Tesla charging stations exclusively use Tesla's charging plug, the North American Charging System (NACS). Until 2025, no other cars came equipped for NACS plugs and instead were equipped for either CCS (also called J1772 combo) or

[2] I provide a table with more information on the three charging levels in Table 5.

[3] There are exceptions to this, but most stations are operated by networks and not franchisees. This is a notable difference from the retail gasoline market.

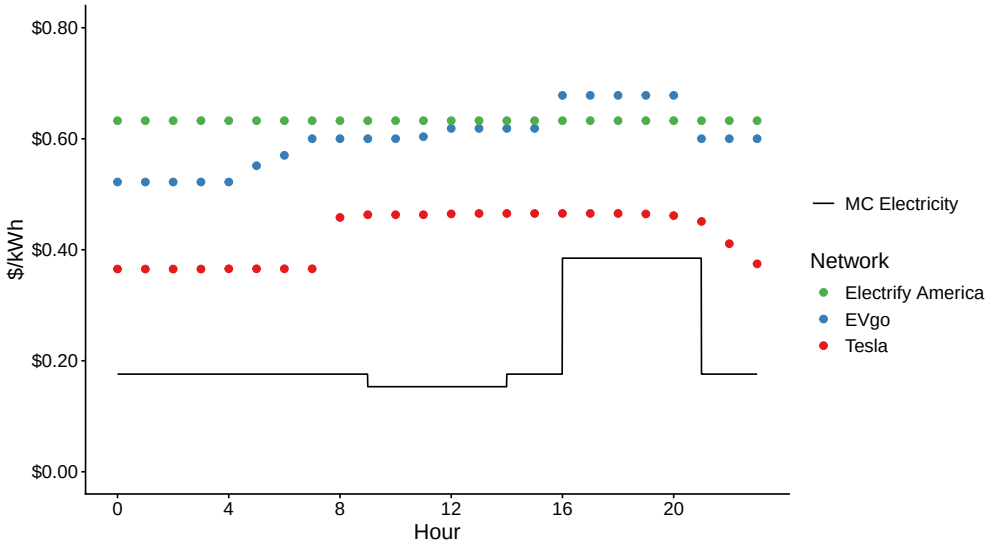
[4] I define stations as any set of multiple chargers operated by a network in a single location.

[5] Another source of charging station information is PlugShare, a web and mobile application that is the largest public aggregator of EV charger information. Neither Tesla nor Electrify America shares prices on PlugShare.

[6] For more on the history of charging standard differences and the consequences for EV adoption, see Li (2023).

CHAdeMO plugs. Further, other stations did not offer NACS charging until 2025, but have begun to incorporate NACS plugs in new stations and retrofit some existing stations with NACS plugs. Automakers sell adapters to allow owners to charge at stations with different charging standards, but development of NACS adapters has been slow and they did not become readily available until the second half of 2025.

Markups above marginal opearting costs are high in the fast charging market. Figure 1 shows networks' average hourly prices at DCFC stations served by PG&E during October 2025. The black line on the figure represents the marginal cost of one kWh of electricity for stations on the most commonly used electricity tariffs during this period.[7] The quantity-weighted average markup above marginal operating cost from October 2024 to October 2025 was 109%, which is an order of magnitude greater than average markups in the retail gasoline industry.[8]



Notes: Data is for October 2025. Prices are averaged across all stations operated by each network in PG&E's service area. "MC Electricity" is the per kWh electricity cost under the most electricity rates most frequently used by fast charging stations.

Figure 1: Average Hourly Prices and Marginal Costs at DCFC Stations in PG&E Service Area

[7]Smaller stations were more likely to use the lower capacity tariffs, which charged slightly higher prices per kWh that were within \$0.02/kWh of the black line at all hours.

[8]Providing charging does not require inputs beyond electricity, so I consider the marginal operating cost to be the cost of purchasing one kWh. There are other fixed operating and maintenance costs such as maintenance and monthly capacity fees that I defer to Section 7.

3 Data

3.1 Station Characteristics

I collect a list of public fast charging stations and characteristics for those stations from the Alternative Fueling Station Location API run by the Department of Energy's Alternative Fuels Data Center (AFDC). The data include station address, geographic location, number of fast chargers, and charging standards (NACS, CCS, and ChaDeMo) available at public charging stations. I use this station list as the foundation for building weekly panels of station prices, charging volumes, and quality.

3.2 Price Data

Each week from October 2024 through May 2026, a team of research assistants and I collected hourly prices for all Tesla, EVgo, and Electrify America DCFC stations in California. Data comes from each company's mobile application and PlugShare, a third party aggregator of charging station data, which posts EVgo prices. The list of stations for a week comes from querying the Alternative Fueling Station Location API for a station list each Saturday. We collected prices for Tesla, EVgo, and Electrify America stations on the station list between Saturday and the following Friday [9] My recorded weekly prices at a station will not capture price changes that occured during that week if the changes occurred after data was collected.

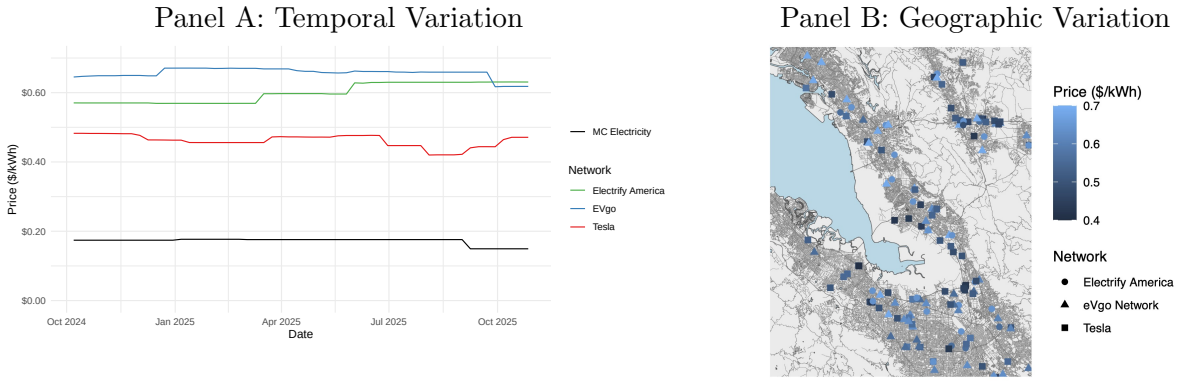
Networks do not necessarily offer the same price to all consumers. EVgo and Electrify America offer subscription plans that lower volumetric prices in return for a monthly fee. Tesla has only opened some of its stations to drivers of non-Teslas, but at those stations prices are 40% higher for charging non-Tesla vehicles unless the consumer subscribes to a monthly plan they offer.[10] I use the default prices from each network (or clarify which set of prices I am referencing). For Electrify America and EVgo, these are the prices that drivers without a paid monthly subscription pay. For Tesla, this can be viewed either as the prices for a Tesla driver or for a non-Tesla driver who purchases a monthly subscription. More than half of EVs registered in California are Teslas, so one can interpret the default price as the price the median or modal driver in California faces. I do not include each network's fees for idling at the station after charging or Tesla's congestion fee for charging above 80%.

In total, the data include 2.6 million station-week-hour prices across 1,407 out of the 1,414 charging stations operated by Tesla, EVgo, and Electrify America on the AFDC's list.

[9] I provide a more detailed explanation of the process in Appendix B.1.

[10] Some Tesla drivers may also face this higher price if they purchased a used Tesla.

Figure 2 shows price variation over time and across geography. Networks change prices rarely, so most price variation comes from within-day and across-station price variation. Networks do not change prices at the same time or in the same direction, which is suggestive that pricing is not collusive. There is spatial variation in prices across a network's stations. Tesla and EVgo predominantly used time-of-use pricing (where prices vary throughout the day) for most of the sample period while Electrify America used a single price at most stations. In November of 2025, Tesla moved to demand-based pricing at most of its stations. These stations switched from prescheduled time of use prices to a high, medium, or low price based on the amount of current use at the station. I cannot view hourly prices at these stations and drop all station-weeks using demand-based pricing from analysis and limit my demand estimation to the period preceding this change.



Notes: Both panels show 2 PM charging prices for stations served by PG&E. Panel A presents average weekly price by network. Panel B shows average price at each of a subset of stations during October 2025.

Figure 2: Variation in 2 PM Charging Prices

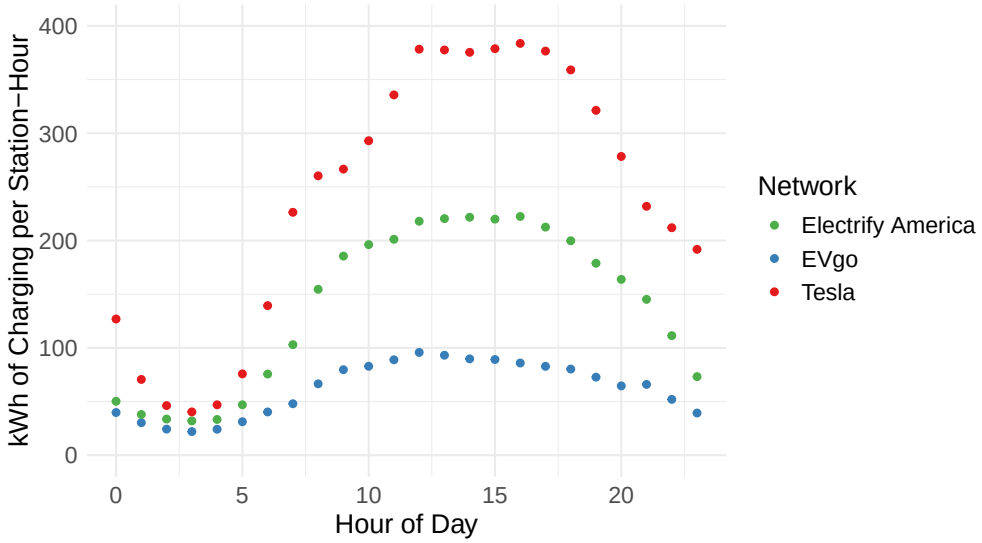
3.3 Electricity Utilization and Billing Data

I obtain station-level electricity consumption and billing data from Pacific Gas & Electric Company (PG&E), the largest electric utility in California and one of the largest in the United States. The data includes both meter-level electricity consumption measured in kWh and rate payer category from 2022 through 2025. The sample includes fast charging stations that PG&E identified in their data or matched to addresses from the AFDC station list.

I convert the raw electricity usage data into an approximation of the amount of electricity used for charging each hour. The raw meter data incorporates all metered electricity consumption at the location, which includes electricity for payment systems and lights. I attempt to remove non-charging electricity consumption by subtracting the lowest observed

electricity use for each station. I classify the remainder as electricity used for charging.[11]

I plot the average hourly use per station by network in Figure 3. Tesla has larger stations than Electrify America and EVgo (see Table 1), and Tesla and Electrify America chargers generally have higher charging speeds. I plot use per charger in Figure 15, which switches the rank order of Tesla and Electrify America. It is also clear that charging activity increases throughout the morning, peaks in the afternoon, and decreases in the evening. This is notably different from grid-wide electricity use which peaks in the evening as individuals return from work and use appliances.



Notes: Data is for October 2025. Electricity utilization is averaged across all stations operated by each network that are served by PG&E.

Figure 3: Average Hourly Electricity Use at DCFC Stations served by PG&E

Using the rate payer category and historical rates, I can determine the marginal cost of one kWh for each station at each hour of the day. Unfortunately, I cannot view the capacity fees paid by stations each month. The PG&E data allow me to measure hourly charging quantities and marginal operating costs at 455 stations that I can identify in the PG&E data and do not share a meter with another station.

3.4 Station Quality Data

My last data source is a measure of quality from Paren, a company that collects charging station data. I use Paren's Station Health Score, which is a weekly index of station quality. The score is calculated using a proprietary algorithm that incorporates a charger's uptime

[11] I document this process more thoroughly in Appendix B.2.

and share of charging session attempts that were successful during the past week. A Station Health Score (referred to as "health index" for the remainder of the paper) of 100 during a week would indicate that a station was operational the whole week and all customers that attempted to initiate a charge at the station did so successfully. Paren's data covers all EVgo and Electrify America stations throughout the sample period and the majority of Tesla stations in 2025.[12] As part of my data use agreement, I do not present quality results disaggregated by network.

Paren began collecting price data in 2025. I match my station-week-hour price data with their price data for September 2025 to assess the accuracy of my data. I find that more than 99% of station week-hour prices match across the two datasets, providing support for the accuracy of my price data.

Table 1: Summary Statistics

	Tesla	EVgo	Electrify America	Other
All California				
Stations	663	454	297	1014
Plugs per Station	16.4	4.9	4.9	5.2
Stations within 5 min.	2.4	2.7	2.1	2.6
Health Index	96.0	80.0	92.8	74.7
Price Data				
Stations	657	454	296	0
Plugs per Station	16.4	4.9	4.9	
Stations within 5 min.	2.4	2.7	2.1	
Health Index	96.1	80.0	92.8	
Price	\$0.43	\$0.59	\$0.61	
Electricity Use Data				
Stations	178	125	90	62
Plugs per Station	15.1	5.5	4.5	6.0
Stations within 5 min.	2.1	3.0	1.9	2.9
Health Index	97.4	81.6	91.5	82.2
Price	\$0.44	\$0.58	\$0.61	
Daily kWh	5761.0	1276.5	2478.7	1060.2

4 Conceptual Framework

4.1 A Stylized Model of Product Differentiation

Markups are an inherent feature of industries with fixed costs and product differentiation. In Bertrand competition, product differentiation allows firms to have price setting power. Only if products are perfect substitutes are there no markups. The more differentiated

[12]Paren initially had limited coverage of Tesla stations, but expanded their Tesla station data in 2025.

products are, the less each firm must consider the actions of the others. In the extreme case, consumers do not substitute between goods and firms act as monopolists. I simplify the model of monopolist competition from Perloff and Salop (1985) to provide a stylized model that illustrates how various features of the charging market contribute to markups.

Consider a consumer that must choose where to charge among n stations indexed by i. The consumer's preference for charging at each station is given by the vector $\boldsymbol{\theta} = (\theta_1, ..., \theta_n)$ where θ_i includes preferences over nearby amenities (such as grocery stores, restaurants, etc.), charging network, charging standard, and travel costs. The vector of prices at each station is $\boldsymbol{p} = (p_1, ..., p_n)$. The consumer chooses to charge at the station that maximizes net surplus $s_i = \theta_i - p_i$ for $i \in 1, ...n$.[13]

For simplicity, let all consumers' preferences for stations be independent and identically distributed $\theta_i \sim U[-1, 1]$. The cumulative distribution function (CDF) is $G(\theta_i) = \frac{\theta_i + 1}{2}$. The probability that a consumer prefers station i to station j is

$$\Pr(s_i \geq s_j) = \Pr(\theta_j \leq \theta_i + p_j - p_i) = G(\theta_i + p_j - p_i) = \frac{\theta_i + p_j - p_i + 1}{2}. \tag{1}$$

Aggregating across L consumers in the market yields station quantities for a given vector of prices, $Q_i(\boldsymbol{p})$. I calculate the Nash-Bertrand equilibrium price:

$$p(n) = c + \frac{L/n}{L/2} = c + \frac{2}{n} \tag{2}$$

where price equals marginal cost c plus a markup inversely proportional to the number of firms in the market, n. I provide a proof that $p(n)$ is the profit maximizing price in Appendix C.

The model has three key implications for the fast charging market. First, station entry increases competition and reduces prices and markups. As n increases, the markup term $\frac{2}{n}$ decreases and p decreases.[14] If firms compete on prices, then increasing the number of firms in the market leads to lower prices. Second, product differentiation sustains markups at equilibrium. Third, preference strength and product differentiation impact equilibrium markups. To see this, let $\theta = \beta\theta_0$ where β is a measure of preference strength and $\theta_0 \sim U[-1, 1]$.

[13] For simplicity, I have omitted the outside option of not charging at any station. Adding in the outside option sets an additional restriction that $s_i \geq \bar{v}$ for the consumer to choose station i. In Section 6, I incorporate the outside option in a heterogeneous agents model.

[14] This implication is reliant on the uniform distribution of θ. I use the uniform distribution for my stylized model to make dynamics more clear, but under other distributions, there is no guarantee that each incremental increase of n will lower the equilibrium price, just that price will converge as $n \to \infty$. However, n is restricted to a finite size for $K > 0$, so without choosing a distribution, it is only possible to show that price converges to the equilibrium for all distributions if $K = 0$.

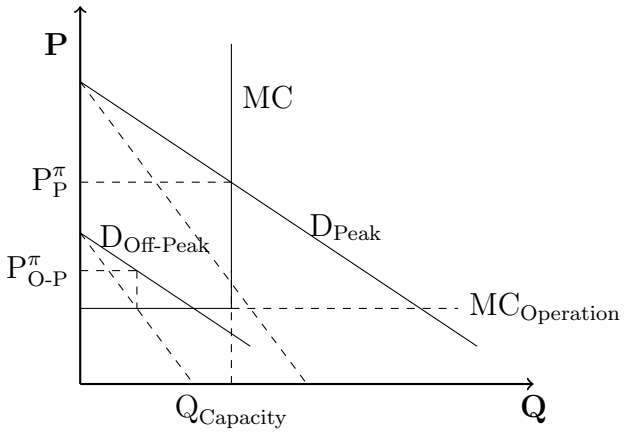


Figure 4: Short Run Efficient Pricing with Capacity Constraints

Then $p(n) = c + \frac{2\beta}{n}$ and the markup is proportional to preference strength. As preferences for charging networks or station amenities increase, equilibrium markups also increase. In the charging market, preferences are likely to be correlated with product characteristics, so greater product differentiation leads to greater markups.

4.1.1 Capacity Constraints

In the stylized model each station sells a single product, but both demand and marginal costs vary throughout the day as shown in Figure 3 and Figure 1, respectively. In the charging market, networks set station-hour-specific prices and customers choose where and when to charge. If all chargers at a station are full, customers are unable to charge there. This is a serious concern for networks. Even though none of the three charging networks shares its prices on Google Maps, both Tesla and Electrify America share the number of available chargers at each station, and all three show this information in their own mobile applications. Further, Tesla switched to demand-based pricing at most of its California stations in November of 2025. Under the new system, prices fluctuate based on the amount of available chargers at a station rather than time of day.[15]

In the short run, networks cannot increase their supply; building new stations or expanding existing stations takes months. At a station's maximum capacity, the marginal cost becomes vertical. In Figure 5, I illustrate the simple case where a station faces different demand in two periods. In both periods, the profit maximizing price is determined by the intersection of the marginal revenue and marginal cost curves.

Peak-period marginal revenue intersects marginal cost on the vertical portion of the

[15]The new pricing system (called "live pricing") sets the price at one of three predetermined prices based on the number of available chargers.

marginal cost curve.[16] This intersection occurs at a lower quantity than the intersection of peak-period marginal revenue and $MC_{Operation}$, so the capacity constraint binds during the peak period and the profit maximizing price, P_P^{π}, is higher than it would be without the constraint. When the capacity constraint binds, there is no markup over the constrained marginal cost, which is undefined, but there is a larger markup over marginal operating cost than there would be in the absence of the constraint.

4.1.2 Long-Run Markups

In the long run, networks choose to build more stations as long as the expected marginal surplus from each station exceeds fixed cost K. In the stylized model stations choose to enter until $p(n) = \frac{nK}{L} + c$, and markups persist even in the long-run equilibrium. In many other markets with fixed costs, such as electricity generation, price equals marginal cost at equilibrium. An upward sloping marginal cost curve allows firms to generate surplus, and the marginal firm generates surplus equal to fixed costs. However, in the fast charging market stations face a constant marginal cost, the cost of one kWh of electricity. In the long-run equilibrium, there will still be markups if marginal costs are constant and $K > 0$.

4.2 Efficient Pricing

4.2.1 Capacity Constraints

In order to understand whether markups are high, it is helpful to establish a benchmark. A benevolent social planner might still set prices that include markups above marginal operating costs, so comparing markups to zero or the long-run competitive level might overestimate the impact of short-run profit maximization on markups. In Figure 5, I return to the two-period example from Figure 4, now from the perspective of the social planner. Short run efficient prices are those at the intersection of the demand and marginal cost curves.[17]

During the peak-period, the socially efficient price P_P^* is equal to the profit maximizing

[16] It is also possible for demand to intersect marginal cost on the vertical portion and marginal revenue to intersect marginal cost on the horizontal portion. This case is the same as the off-peak demand case. The station would price based on the quantity at which marginal revenue equals marginal cost, which is less than the quantity at which demand and marginal cost intersect.

[17] Readers may have noticed I have dropped external costs and moved from social marginal cost to private marginal cost. Borenstein and Bushnell (2022) focus on the importance of considering external costs in determining efficient electricity rates. They show that electricity rates in California are well above social marginal cost, which means the private cost that charging stations face is greater than the social marginal cost of producing that electricity. Therefore, $P_{O\text{-}P}^*$ does not maximize social surplus, but rather maximizes surplus contingent on the restriction that the charging network not take losses on each unit of charging. Truly maximizing social surplus would require either PG&E to lower the per-unit cost of electricity or for the state to pay per kWh subsidies to charging stations.

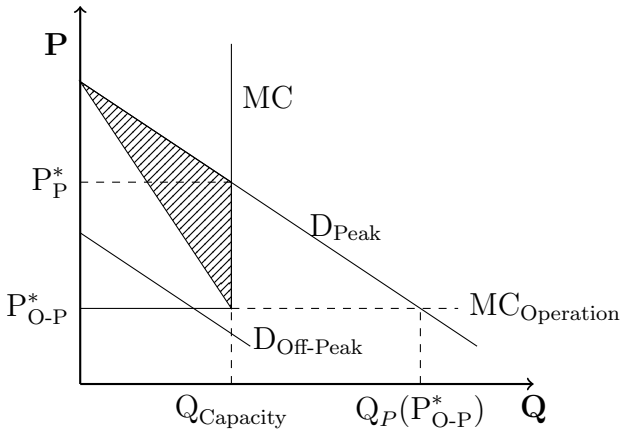


Figure 5: Short Run Efficient Pricing with Capacity Constraints

price P^{π}_{P} shown in Figure 4. Charging a price of $P^*_{O\text{-}P} = MC_{Operation}$ would lead to excess demand, and there is no guarantee that charging will be allocated to the consumers who receive the greatest benefit. To clarify this point, suppose that charging is randomly allocated across all consumers willing to pay $P^*_{O\text{-}P}$ during the peak-demand period. The expected willingness to pay of any consumer who receives charging is halfway between D_{Peak}'s P-intercept and $P^*_{O\text{-}P}$. The shaded region represents the expected deadweight loss from this random allocation, which does not increase quantity and leads to a lower average willingness to pay (relative to the market clearing price, P^*_P) among consumers who are able to charge.

This example shows that profit maximizing markups can be efficient. When demand is sufficiently high, the social planner and profit maximizing firm choose the same price. P^*_P equates quantity supplied and demanded to clear the market. It is the lowest price that eliminates the deadweight loss from misallocation, and the highest price that eliminates deadweight loss from an inefficiently low charging quantity.

In the long run, capacity constraints still bind. The social planner chooses to add capacity if the additional social surplus exceeds the fixed costs. In Figure 5, adding capacity would only provide social surplus during the peak period, and generates less surplus during the peak period than existing capacity. The decreasing marginal surplus implies an equilibrium with capacity constraints and prices above the marginal operating cost.

4.2.2 Fixed Cost Recovery

The social planner decides to build stations that increase social surplus even if they generate negative profits. No profit maximizing firm would choose to build a station that generates negative profits, so a better benchmark may be to require that stations recover fixed costs. In this case, the social planner sets prices but firms choose whether to build

stations. Firms only build stations if they expect to earn positive profits, so the social planner must set prices high enough to justify entry.

In the single-product case, a social planner would set a station's price equal to the average cost (for a station that generates positive social surplus). That is, $p_i = c + \frac{K}{Q_i}$, the marginal cost plus a markup to recover fixed cost. Setting price above marginal cost creates deadweight loss from reducing consumption below the optimal amount.[18] The amount of deadweight loss is a function of the size of the markup and demand elasticity. Holding the markup constant, as demand becomes more elastic, consumption falls further from the efficient quantity.

In the charging station context, the social planner sets prices for all 24 hours of the day and can apply different markups to each. It is more efficient to recover fixed costs by applying larger markups when demand is most inelastic because consumption will be the least impacted (Ramsey, 1927). The set of linear prices that minimize deadweight loss while recovering fixed costs, often called "Ramsey prices," sets markups proportional to elasticities of demand across products (Boiteux, 1960; Baumol and Bradford, 1970).

A more efficient alternative is to use two-part tariffs (Feldstein, 1972; Littlechild, 1975). A network would set prices closer to the social surplus maximizing prices and extract consumer surplus by requiring subscription fees or raising car prices if they are vertically integrated. Electrify America, EVgo, and Tesla all offer subscriptions and are at least somewhat vertically integrated – Tesla sells EVs in addition to charging, Electrify America is owned by Volkswagen, an EV manufacturer, and EVgo has partnered with GM, another EV manufacturer – so each has the ability to use two-part tariffs. While I omitted two-part tariffs from the profit-maximizing firm's considerations in Section 4.1, they increase a monopolist's profits above those using linear prices (Oi, 1971). Networks have the ability and potential incentive to use two-part tariffs. Doing so would likely increase social surplus relative to the Ramsey prices a social planner might set.

In a competitive market, the number of firms, capacity constraints, and fixed costs can all lead to markups. Alternatively, extreme product differentiation or rigid consumer preferences could reduce substitution and allow networks to act as monopolists. Section 5 addresses whether networks compete on prices and whether market entry reduces prices. In Section 7, I return to capacity constraints and fixed costs as possible explanations and compare observed prices to those that the efficient prices initially discussed in this section.

[18] I am ignoring capacity constraints for the time being – I will combine capacity constraints with fixed cost recovery in Section 7.

5 Station Entry

In this section, I estimate the impact of station entry on incumbent prices, quality, and charging volumes. This combination of outcomes allows me to test for different forms of competition. If stations respond to competing-network entry by lowering prices, this aligns with the model of Bertrand competition amongst partially differentiated stations. If they do not lower prices but do improve quality, this would suggest stations are competing on quality but not price (they could also compete on both).

There are multiple reasons stations might not adjust prices or quality in response to competing-network entry. First, if there is no response and incumbent stations sell lower quantities following competitor entry, this indicates that networks do not profit maximize with linear prices. One explanation would be that they use two-part tariffs and have already set prices as low as maximizes social surplus. Another explanation is that supply-side frictions lead to sub-optimal firm behavior (Arcidiacono et al., 2020). Second, if there is no change in quantities, prices, or quality, then networks do not compete at the station level because consumers do not substitute between networks. Such a result would indicate that either consumers do not view different networks as substitutes or that search costs or other demand-side frictions deter cross-network substitution.

5.1 Empirical Design

I consider the effect of station entry on incumbent charging quantities, prices, and quality. I limit my sample to incumbent stations within a five-minute drive of the entrant to focus on dynamics between stations that one would expect to be substitutes.[19] I measure responses to market entry by exploiting the exact timing of entry. Networks choose new locations based on market conditions, but the licensing and infrastructure upgrade process takes several months and is subject to a large amount of variation in how long it takes. Once operational, stations have no reason to delay opening as operational costs are below their expected revenues. This means that the exact timing of entry is determined by administrative and technical factors rather than market conditions. My identifying assumption, that stations do not control the exact timing of entry, is the same that Arcidiacono et al. (2020), Davis, McRae and Seira (2025), Taylor and Muehlegger (2025), and others use.

My primary empirical approach is a difference-in-differences event study model. Station construction is public, so a key threat to causal identification is anticipation. If stations lower prices in anticipation of station entry, then a standard difference-in-differences approach that

[19] I explain the role of distance in Section 5.2.6 and provide results using a 10-minute threshold in Appendix A.

averages pre-period prices would underestimate the counterfactual price in the absence of entry and bias the difference-in-differences estimate. I use an event study specification to show trends during the twelve weeks prior to station opening and disaggregate results by the number of weeks from treatment. This approach also has the benefit of showing how long it takes for quantities and prices to adjust.

I match treated stations to untreated stations to create a counterfactual for outcomes in the absence of openings. I designate untreated stations as those that did not have an opening within a 10-minute drive within 24 weeks of the treatment opening. I limit matching to untreated stations operated by the same charging network, in the same core-based statistical area (CBSA), and with a similar number of nearby stations.[20] The goal of this approach is to compare treated stations to stations that were affected by the same market trends and faced a similar amount of competition prior to the entry event. The k counterfactual stations for each treated station are given a weight of $\frac{1}{k}$ to ensure each treatment event is weighted equally.

I provide a visual representation of this process in Figure 6. In this example, a new Tesla station (red circle) opens within a five-minute drive (depicted by the red line) of a Tesla (green circle), Electrify America (blue square), and EVgo (orange triangle) station. Each of those stations is matched to all the corresponding stations that are the same color and shape as the treated station. Those stations are all part of the same charging network, in the San Diego-Carlsbad CBSA, and near a similar (within one) number of stations. The map illustrates why I require nearly the same number of nearby stations for control stations. When electricity rates fall and marginal costs decrease, EVgo might lower prices at the orange triangle stations but leave prices unchanged where it has little competition, such as its station on Coronado Island, a black triangle towards the bottom left of the figure.

The event study runs from twelve weeks before entry to twelve weeks after entry. I avoid overlapping treatment by restricting to only entrances near an incumbent without another nearby entry event within 24 weeks. I consider incumbent electricity use, price, and health index as outcomes and run regressions separately for each network for quantity and price in case driver behavior and incumbent strategies vary by network.[21] I estimate the equation:

[20] I require the number of nearby stations at matched stations to be within one when the treated station has four or fewer nearby stations, within two when the treated station has five to nine nearby stations, or within three when the treated station has 10 or more nearby stations.

[21] For regressions with health index as the outcome variable, I present results aggregated across all networks due to my data sharing agreement.

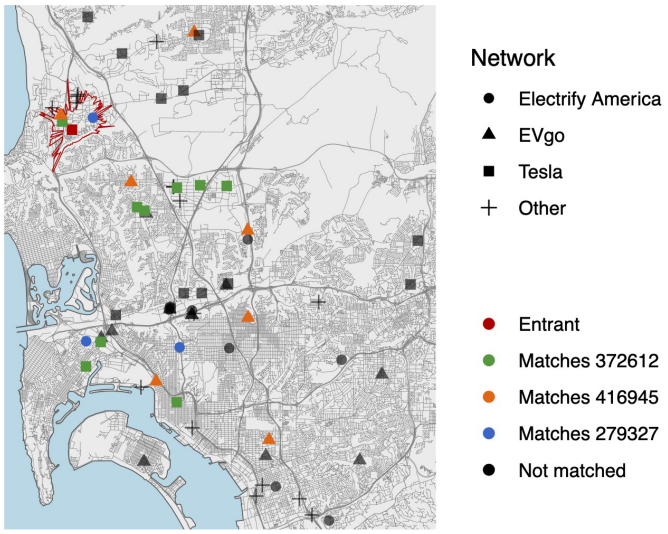


Figure 6: Visual Example of Matching Process

$$y_{ihw} = \sum_{\tau=-12}^{\tau=12} \beta_{\tau d} \mathbf{1}[\text{EventWeek}_{iw} = \tau] \times \mathbf{1}[\text{Treated}_i] + \mu_{ih} + \gamma_{n(i)w} + \lambda_{u(i)m(w)} + \varepsilon_{ihw} \quad (3)$$

where τ represents event week and $\mathbf{1}[\text{Treated}_i]$ is a dummy variable equal to one for incumbents exposed to entrants and zero for the counterfactual group. I estimate station price and usage at the station-week-hour level to account for different prices and charging over the course of the day. I include station-hour fixed effects, μ_{ih}, to control for time-invariant station characteristics, network-week fixed effects, $\gamma_{n(i)w}$, to control for network pricing behavior over time, and utility-month of sample fixed effects, $\lambda_{u(i)m(w)}$, to capture changes in marginal costs over time. I estimate the impacts of opening on health index at the staion-week level and therefore include station fixed effects instead of station-hour fixed effects.

I apply two-way clustered standard errors at the CBSA and year-month level, because the probability of treatment varies across CBSAs and time. I do not interact CBSA with year-month because the error for one CBSA in a given month is likely correlated with the error the previous month (Colin Cameron and Miller, 2015). The total number of clusters ranges from 51 to 59 across analyses.

I elect to include already treated stations in my analysis. This is because most stations that have a nearby competitor have already been treated (even if treatment was before the study period) and I want to understand the effect in both areas where there are already multiple stations (in addition to those with few stations). I set a 24 week threshold between entry events to ensure that no station-week in my analysis is treated by multiple entries. In

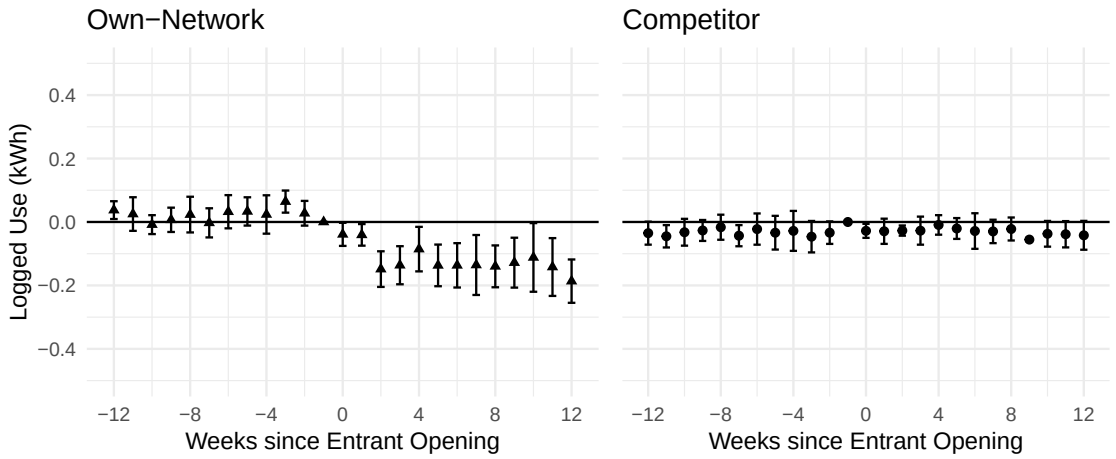
these cases, I drop all entry events to ensure the pre- and post-periods are not contaminated. Figure 16 shows that limiting to just the first entry would cut my sample size by 25% and that station exit is incredibly rare during this time.

In addition to the event study models, I also estimate a standard difference-in-differences model. This model uses the same data but groups the twelve pre-entry weeks into a single pre-period group and the week of entry and following twelve weeks into a single post-period group. The difference-in-differences estimator recovers the change in the difference between the treatment and control group from before to after entry.

5.2 Results

5.2.1 Charging Quantities

Market entry has asymmetric effects on own- and competing-network stations. Figure 7 shows that charging at existing stations decreased following own-network entry but remains unchanged following competing network entry. The decrease from own-network entry occurs shortly after station entry and is sustained for the event study's twelve week window at around a 14% decrease from just before entry. Figure 17 shows the own-network effect is driven by Tesla stations and Figure 18 shows the effect of competitor entry on charging is null across all networks. I provide results from the difference-in-differences model for these and the following event study analyses in Table 2.

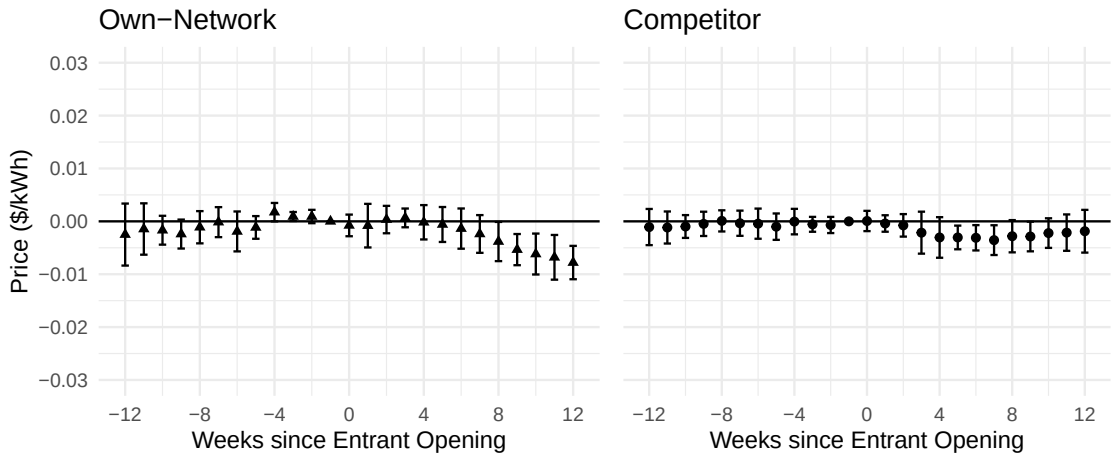


Notes: Data is for total electricity consumption at each station and spans January 2022 - December 2025.

Figure 7: Event Study Estimates of Own-Network Station Entry within Five Minutes on Incumbent Hourly Electricity Consumption

5.2.2 Prices

Price responses generally align with changes in charging volumes. Figure 8 shows no anticipatory price changes in the twelve weeks before a station's opening. After station entry, prices fall slightly at own-network stations but not competing stations. Again, the own-network effect comes from Tesla stations and the non-response to competitor entry is consistent across networks (see Figure 19 and Figure 20).



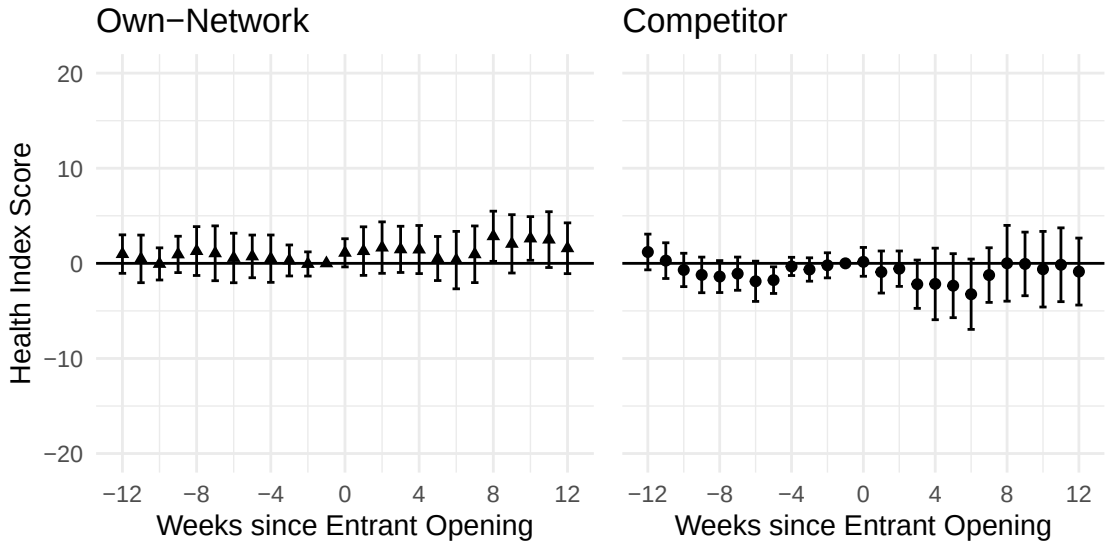
Notes: Data is for station-week-hour prices and spans October 2024 - May 2026.

Figure 8: Event Study Estimates of Own-Network Station Entry within Five Minutes on Incumbent Price

The use results showed that Tesla's new stations cannibalize some demand from any nearby incumbents, shifting demand down and leading to a lower optimal price. The gradual decrease does not reflect Tesla changing prices repeatedly, but rather that Tesla rarely changes prices, so the price change may lag the demand change by a couple weeks or months depending on when the network next updates prices. Similarly, stations experience no decrease in demand following competitor entry and therefore have no incentive to lower prices. The difference-in-differences results in Table 2 suggest I can rule out average competitor price responses greater than 0.4 cents and network-specific prices responses greater than 0.7 cents. Using the average price and marginal electricity cost for the stations in the PG&E data, the confidence interval bound of 0.4 cents maps to 1.5% of price minus marginal operating cost margins. Therefore I can rule out price decreases in the realm of the 6-7% estimated for the retail gasoline industry (Taylor and Muehlegger, 2025; Davis, McRae and Seira, 2025).

5.2.3 Quality

I also find no evidence that incumbent stations improve quality in response to greater competition. Figure 9 shows the effect of station entry on own-network and competing station health index score. Taken together, the price and quality results show stations do not appear to respond to compeitor entry. This is a rational decision – I detect no change in demand for stations following competitor entry, so the optimal prices and quality after entry should be the same as prior to entry. It is still possible that stations compete on other measures, or that competition is at the network level – networks make decisions on prices and quality that affect all stations in an area at once. I explore network-level pricing in Section 7.



Notes: Data is for station-week health index score and spans October 2024 - May 2026.

Figure 9: Event Study Estimates of Station Entry within Five Minutes on Station Quality

My results indicate that networks do not compete at the station level on prices or quality. There is no incentive for them to do so, because consumers do not substitute across network to new stations. This combination of results suggests that consumers choose between the stations in a network rather than all stations. It is possible that networks or stations are highly differentiated (due to strong network or nearby amenity preferences) or that demand-side frictions reduce cross-network substitution and deter competition. This result does not rule out that networks compete at a broader geographic level such as across the CBSA or state. It is possible that networks change prices in a competitive way periodically at all stations, that consumers choose between networks based on prices or quality, and that once they've chosen a network, consumers do not search outside of their network. Panel A of

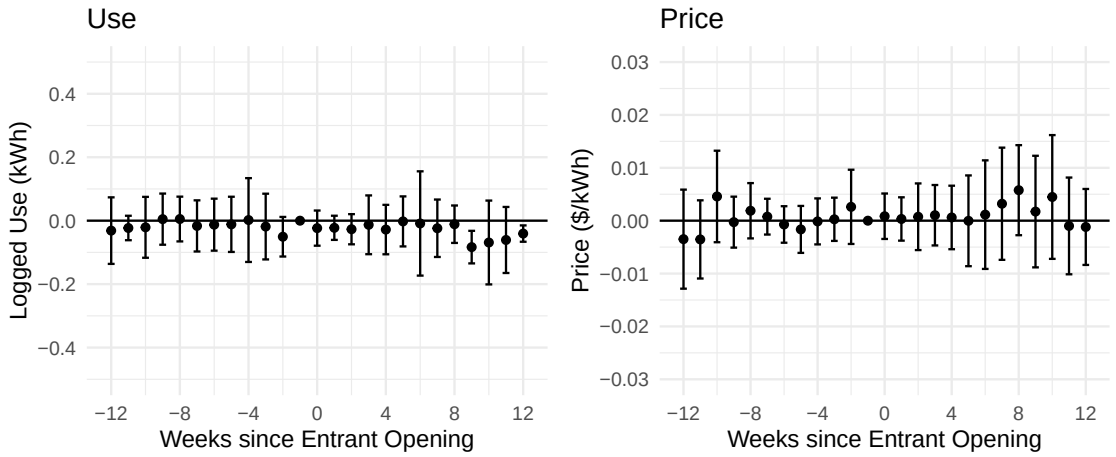
Figure 2 shows that networks do not change prices in response to other networks' price changes, so if networks do compete at a broader geographic level, such price competition is not dynamic.

5.2.4 Charging Standards

One possible explanation for why I detect no cross-network substitution is that different networks use different charging standards. Almost all cars are only equipped to charge with one charging standard, so drivers must either go to a station with that standard or own the requisite adapter. Prior to 2025, all non-Tesla stations used CCS, CHAdeMO, or both, while Tesla exclusively used NACS. Since 2025, non-Tesla networks have begun to build new stations with NACS plugs and retrofit old stations to include NACS plugs (though the latter is not common). Non-Tesla manufacturers also began to make NACS adapters available to drivers in 2025, though they were in short supply at the start of the year. My use data cover 2022 through 2025, so a preference for stations with NACS chargers could appear as a network preference.

I use three approaches to evaluate whether the lack of substitution across networks reflects preferences for charging standards rather than networks. First, I run the same usage event study for competitor entry as in Section 5.2.1 but limit to just entrants using the same charging standard as the incumbent. The changes in charging standards and converter accessibility in 2025 motivate two additional experiments. The first is to limit the window of analysis to 2025 onwards. If non-Tesla networks begin competing with Tesla for customers in 2025 now that more non-Tesla stations had NACS plugs and more non-Tesla drivers had NACS adapters, focusing on just the period with competition should reveal an impact of competitor station openings. Second, my weekly panel of station information allows me to determine when networks retrofitted existing stations to add NACS chargers. I use the same event study approach to determine whether charging volumes or prices at nearby existing Tesla stations change following addition of NACS chargers.

Figure 10 shows that there is no substitution across stations from different networks that have the same charging standard and stations still do not lower prices. I provided results by network for use and price in Figure 21 and Figure 22, respectively. Additionally, there is no substitution from Tesla stations to non-Tesla stations when the non-Tesla stations add NACS chargers (see Figure 23) or to new non-Tesla stations in 2025 when these stations are potentially better substitutes (see Figure 24). These results suggest it is network rather than charging standard differences that drive the lack of substitution across networks. One potential explanation is that drivers do not consider all possible stations with the charging plug type they need, but instead open one network's application to look for nearby stations.



Notes: Data is for total hourly electricity consumption at each station and spans January 2022 - December 2025.

Figure 10: Event Study Estimates of Entry by Competing Station with Same Charging Standard within Five Minutes

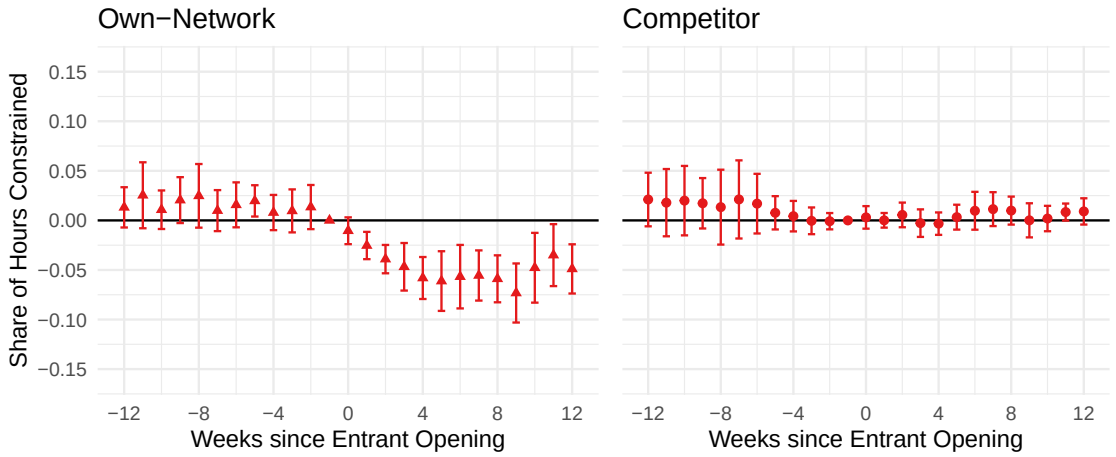
5.2.5 Capacity

A notable difference between the retail gasoline literature and my results is that Tesla lowers their prices after they open a station near one of their own stations. One potential explanation is that the new station cannibalizes demand, leading to a downward shift in demand at the incumbent stations. This shift leads to a decrease in optimal price. It is also possible that capacity constraints explain part of the price effect.

Stations are limited in how many cars they can charge and how much electricity they can discharge at once. When all chargers at a station are full, the marginal cost curve becomes vertical as the station would need to add another charger to increase quantity, but cannot do so in the short run. Even when the station is not completely full, an additional vehicle may slow charging speeds (because technical constraints restrict the maximum electricity a station can discharge) or deter potential customers from driving to the station if they use a mobile application to check how full the station is before driving there. This could be modeled as an increase in the marginal cost of providing one kWh above the marginal operating cost. If Tesla entrants relieve capacity constraints, this would reduce marginal costs at the incumbent station by eliminating periods where capacity constraints increase marginal costs. This may explain why prices fall at Tesla stations following nearby Tesla entry and why the network chooses to build stations that will cannibalize charging from existing stations.

I use the same event study approach from the prior entry analyses to test whether entry relieves capacity constraints. Ideally I would define a station as capacity constrained during

any hour where the station must reduce the amount of electricity drawn by each charger or there are ever no chargers are available. I cannot observe either in my data, so I use a rough measure of how close electricity discharge is during an hour to the maximum amount of electricity the station can discharge. I calculate the observed electricity usage during any given hour during the station's history and designate an hour as potentially capacity constrained (at or near capacity) if electricity consumption during that hour is at least 80% of a benchmark for the maximum feasible amount of electricity consumption. I set that benchmark as each station's 98th percentile of electricity consumption during daytime charging hours.[22] Using this approach, I classify 5.4% of station hours as capacity constrained.



Notes: Data is for total hourly electricity consumption at each station and spans January 2022 - December 2025.

Figure 11: Event Study Estimates of Station Entry within Five Minutes of Tesla Station on Share of Hours near Charging Capacity

The events study results in Figure 11 show that Tesla stations exposed to new own-network stations significantly reduce the likelihood of capacity constraints. The probability of a Tesla station being potentially capacity constrained decreases by 6.6 percentage points. This estimate is very large in magnitude – a 47% decrease from the 14 percent of pre-period hours classified as potentially capacity constrained. This suggests stations may lower prices following nearby own-network entry because they are less likely to be capacity constrained.

[22]Charging speeds vary by car model and the car battery's charge level (charging gets notably slower once the battery is more than 80% full – Electrify America doesn't allow charging past 85% at some stations and Tesla charges extrea for cars charging above 80%) so there is no set amount of electricity a fully occupied station would discharge. I therefore use the 98th percentile of charging from 8 AM to 8 PM to rule out hours where cars were exceptionally fast in terms of charging speeds. Changing this threshold up or down makes a slight difference in the share of hours classified as at capacity, but no difference in the relative effect size.

Table 2: Difference-in-Differences Results

	All		Tesla		EVgo		Electrify America
	Competitor	Own	Competitor	Own	Competitor	Own	Competitor
Log Use (kWh/hour)							
Estimate	0.000	-0.143**	0.001	-0.179**	0.014	-0.073*	-0.014
	(0.014)	(0.019)	(0.017)	(0.028)	(0.025)	(0.013)	(0.036)
Baseline mean	4.557	4.652	5.371	5.457	3.624	3.274	4.294
Observations	1,309,200	335,400	478,200	147,000	420,600	117,000	367,200
Price ($/kWh)							
Estimate	-0.002	-0.002	-0.002	-0.005*	-0.001	0.006**	-0.001
	(0.001)	(0.002)	(0.004)	(0.002)	(0.001)	(0.001)	(0.003)
Baseline mean	0.554	0.468	0.415	0.424	0.604	0.590	0.612
Observations	5,108,616	2,007,072	1,512,216	745,056	2,371,512	854,928	1,224,888
Health Index							
Estimate	-0.463	0.948					
	(1.553)	(0.959)					
Baseline mean	88.6	93.1					
Observations	2,998,560	848,208					
Share Hours with Constrained Capacity							
Estimate	-0.004	-0.046**	-0.007	-0.061**	-0.002	-0.013	-0.001
	(0.007)	(0.008)	(0.008)	(0.005)	(0.003)	(0.004)	(0.008)
Baseline mean	0.079	0.109	0.096	0.126	0.052	0.073	0.080
Observations	1,309,200	335,400	478,200	147,000	420,600	117,000	367,200

Notes: $*/**$ Denote significance at $p < 0.05/0.01$ level.

5.2.6 Distance

In the previous models, I limit treatment to the effect of openings within five minutes and compared to similar stations without any openings within ten minutes. In this section, I test this assumption on the distance over which consumers are willing to substitute between stations. I compare stations treated by openings that are 0-5 minutes, 5-10 minutes, and 10-15 minutes away to control stations without any openings within a 15-minute drive. I focus specifically on the effct of Tesla openings on Tesla incumbents as this combination of existing and new stations was the one with the largest magnitude estimates.

I increase sample size for the analysis of the effect on charging quantities by shrinking the analysis window to eight weeks before and after opening. The results in Figure 7 show that the effect of entry on charging quantities is stable after two weeks, so changing the window should not be a concern for the parallel trends requirement. The effect on prices in Figure 8 doesn't materialize until six to eights weeks after opening and continues to grow through the study period, so I do not change the analysis window for the price analysis.

I present results for the effects of own-network openings on incumbent Tesla stations in Figure 25. The effect of new Tesla stations on nearby existing Tesla station charging quantities is largest for the 0-5 minute treatment group, but also negative and significant for

the 5-10 minute treatment group. Even though there is no effect on charging for stations 10-15 minutes away, all three groups decrease prices following entry. This could reflect that rather than lowering prices in response to decreased demand, Tesla is decreasing prices in anticipation of a demand response, but it doesn't materialize for the stations 10-15 minutes away. Because my results suggest that drivers are willing to switch to new stations 5-10 minutes away from an existing station, I conduct all event studies in this section using a 10-minute treatment window and provide those results in the appendix (see Figure 26-Figure 30).

6 Demand Estimation

6.1 Model

The results in Section 5.2 show that consumers do not substitute to new stations from different networks. However, they do not answer whether consumers respond to prices. To answer this question and whether consumers substitute across time of day, I estimate a model of heterogeneous demand. In Section 4.1, I modeled discrete choice among consumers using a simplistic model that limited consumer preferences and the number of products in the market. I now update that model to account for heterogeneous preferences and utilize my extensive data.

EV drivers must regularly choose where and when to recharge their vehicles. Preferences for charging location are likely based on a variety of factors including price, proximity, nearby amenities, time of day, charging speed, and reliability.[23] Importantly, those preferences are likely to be heterogeneous across consumers. Therefore, my model builds on the framework for estimating models of discrete choices across heterogeneous consumers developed by Berry (1994) and Berry, Levinsohn and Pakes (1995).[24]

In each period t in market m, consumers choose when and where to charge their electric

[23] Networks have periodically offered free charging with the purchase of a car, which could lead to some consumers having perfectly inelastic demand with respect to price. Using California registration data, I estimate that the share of EV owners with free charging in 2025 could be 9%. However, I expect that it is notably lower as many of these offers were limited to a relatively low amount of free charging.

[24] I make three assumptions to consider charging quantities in a discrete framework. First, I make the same assumption as Nevo (2001), that consumers do not gain utility from consuming from multiple chargers. Their decisions can be seen as multiple purchase instances that are independent of each other, which avoids the need to model consumption as a continuous variable. Second, I assume that the choice set does not impact the amount that consumers drive. The total amount of gasoline purchased is highly inelastic, which is in line with this assumption: if the price of driving decreases, consumers will still drive the same amount. Third, I assume that consumers have decided on an optimal amount to charge before choosing their charging station. This is similar to Hendel (1999), who models firms buying multiple units of several brands of computers as making the optimal choice on many tasks, which in aggregate yield the observed purchase.

vehicle.[25] Consumer i's indirect utility from charging their vehicle at a certain station during a specific hour, jointly denoted as product j, is given by the equation:

$$u_{ijt} = x_{jt}\beta_{it} - \alpha_{it}p_{jt} + \xi_{jt} + \varepsilon_{ijt} \tag{4}$$

where α_{it} and β_{it} reflect preferences over prices, p_{jt}, and a vector of other observable station characteristics, x_{jt}, respectively. ξ_{jt} captures the utility from station-hour attributes that are observed by consumers and the network but not the econometrician.[26] Indirect utility also includes individual preference shocks ε_{ijt}. I assume ε_{ijt} are i.i.d (across (i, j, t)) draws from a type I extreme value distribution, and are observed by neither the econometrician nor network. This specification yields the mixed Logit model.

The mixed Logit allows preferences to vary across consumers and time. For example, individual i's preferences over price and observable characteristic k are given by the following two equations:

$$\beta_{it}^{(k)} = \beta_0^{(k)} + \beta_\nu^{(k)} x_{jt}^{(k)} \nu_{ik} \tag{5}$$

$$\alpha_{it} = \alpha_0 + \alpha_\nu \nu_{ip} \tag{6}$$

where β_0 and α_0 measure average utility across all consumers and the terms $\beta_\nu^{(k)} x_{jt}^{(k)} \nu_{ik}$ and $\alpha_\nu p_{jt} \nu_{ip}$ capture deviations in preferences from the population average. It is convenient to rewrite the utility as a function of average and individual preferences:

$$u_{ijt} = \delta_{jt} + \mu_{ijt} + \varepsilon_{ijt} \tag{7}$$

where δ_{jt} is the mean utility consumers receive from choosing to charge at station-hour j during period t, and μ_{ijt} is the individual (mean zero) deviation from that mean utility. I decompose the mean utility term into both observed and unobserved components:

[25] No station hour can be in two markets, so I do not include m in notation because it is fully determined by station hour j.

[26] For example, suppose a charging station is located in the parking lot of a grocery store and the grocery store goes out of business. The station may anticipate reduced charging demand and respond by lowering prices. The econometrician would underestimate the price elasticity of demand because they would assume that the change in quantity was purely a function of the price change (if no other characteristics changed).

$$\delta_{jt} = x_{jt}\beta_0 - \alpha_0 p_{jt} + \xi_{jt} \tag{8}$$

where β_0 and α_0 reflect average utility from x_{jt}, a vector of K observable characteristics, and p_{jt}, the price per kWh of charging, respectively. ξ_{jt} captures the impact of unobserved station-hour attributes on utility. The individual deviation can be expressed as:

$$\mu_{ijt} = \sum_{K} \left[\beta_\nu^{(k)} x_{jt}^{(k)} \nu_{ik}\right] - \alpha_\nu p_{jt} \nu_{ip} \tag{9}$$

where K is the vector of observable characteristics. For simplicity, I will instead write $\mu_{ijt} = (x_{jt}, p_{jt}) \cdot \Sigma\nu_{it}$ where Σ is a $(K+1) \times (K+1)$ diagonal matrix with the diagonal $(\alpha_\nu, \beta_\nu^1, ..., \beta_\nu^K)$ and ν_{it} is the vector of individual preference shocks.

Consumer i chooses to charge at station-hour j in period t if $u_{ijt} \geq u_{ikt}$ for all $k \neq j$, $j, k \in m$. One possible k is the outside option, charging at a slower charger or station not in my data, which gives utility $u_{i0t} = \varepsilon_{i0t}$. That is, consumers choose the station-hour that gives them the highest utility of all station-hours in market m during period t, unless this is less than the utility from the outside option.[27] Integrating over all consumer choices using the distribution of consumer preferences, $F(\nu_{it})$, yields the market share for each station-hour:

$$s_{jt} = \sigma_j(\boldsymbol{\delta}_t, x_t, p_t; \Sigma) = \int \frac{\exp(\delta_{jt} + \mu_{ijt})}{1 + \sum_{k=1}^{J} \exp(\delta_{kt} + \mu_{ikt})} dF(\nu_{it}) \tag{10}$$

where $\boldsymbol{\delta}_t = (\delta_{1t}, ..., \delta_{Jt})$, $x_t = (x_{1t}, ..., x_{Jt})$, and $p_t = (p_{1t}, ..., p_{Jt})$.

6.2 Estimation

The goal of estimation is to recover the vector of parameters $\theta = (\alpha_0, \beta_0, \Sigma)$. One could theoretically estimate the parameters by minimizing the distance between the observed and predicted market shares. However, such an approach does not account for the correlation between prices and unobserved product characteristics, ξ_{jt}. To handle potential correlation, I implement the approach used by Berry, Levinsohn and Pakes (1995).

The first step is to recover mean utility, δ_{jt}, for each product and time. The presence of ξ_{jt} means I cannot estimate β_0 and α_0 directly and instead calculate mean utility. Berry

[27] The majority of outside option charging is home charging, but the outside option also includes workplace charging and some charging at fast charging stations I do not observe.

(1994) and Berry, Levinsohn and Pakes (1995) show that one can fix the elements of Σ at predetermined values and invert the market shares to yield $\hat{\delta}_{jt} = \hat{\sigma}_{jt}^{-1}(\hat{s}_{1t}, ..., \hat{s}_{Jt}; \Sigma)$.

The second step is to regress $\hat{\delta}_{jt}$ on x_{jt} and p_{jt} to estimate α_0, β_0 using the equation $\hat{\delta}_{jt} = x_{jt}\beta_0 - \alpha_0 p_{jt} + \xi_{jt}$. However, ξ_{jt} are observed by consumers and the network but not the econometrician. ξ_{jt} likely impact prices and may be correlated with the observed variables x_{jt} (which are assumed to be exogenous), which would bias estimates of α_0, β_0. The solution is to use a vector of exogenous variables, $\boldsymbol{Z}_t$, to build a vector of instruments, z_{jt}, for p_{jt}. One can then estimate an instrumental variables regression to recover $\hat{\alpha}_0, \hat{\beta}_0$ (I explain my instruments in Section 6.2.1).

The estimation procedure explained so far relies on holding Σ fixed while one estimates β_0 and α_0. The identifying assumption for both the instrumental variables regression in the second step and estimation of Σ is $E\left[\xi_{jt}|\boldsymbol{Z}_t\right] = 0$, which implies $E[z_{jt}\xi_{jt}] = 0$ for any z_{jt} constructed from $\boldsymbol{Z}_t$. The empirical analogue for the nth moment is $m^n(\theta) = \hat{E}[z_{jt}^n \xi_{jt}]$. One can stack all the moment conditions to form $\boldsymbol{m}(\theta)$ and create the generalized method of moments (GMM) estimator:

$$\hat{\theta} = \arg\min_{\theta} \boldsymbol{m}(\theta)' W \boldsymbol{m}(\theta) \tag{11}$$

where W is a positive definite weighting matrix with dimension $N \times N$. Conlon and Gortmaker (2020) calculate a GMM objective $q(\Sigma) = \boldsymbol{m}(\theta)' W \boldsymbol{m}(\theta)$ (note that $q(\Sigma)$ does not include $\alpha_0 and \beta_0$, which concentrated out from θ because, given each Σ, they have a closed-form solution). They continute to try new Σ until the gradient of q with respect to Σ is approximately zero and one can conclude that $\boldsymbol{m}(\theta)' W \boldsymbol{m}(\theta)$ has been minimized. At each guess of Σ, they repeat the first and second steps and evaluate $q(\Sigma)$.

Any positive definite W gives a consistent estimate of θ, but only one W provides efficient estimates. The efficient weighting matrix is the inverse of the asymptotic covariance matrix of the population moments. Therefore, in order to get the efficient estimates, one must solve the problem twice. The first solution yields a consistent estimator that is used during the second iteration to recover the efficient estimator.

I conduct the estimation process using the *PyBLP* Python package from Conlon and Gortmaker (2020), which is based on the approach developed by Berry, Levinsohn and Pakes (1995). I estimate the mixed Logit model with markets defined at the week by CBSA level.[28] CBSAs are spatially distinct and there are very few stations that have a nearby station not in their CBSA, so it is unlikely that consumers are substituting across CBSAs. I choose to

[28] I drop CBSAs without at least ten stations, which leaves seven CBSAs and 56 weeks totaling 392 markets.

divide use into weeks to match my price data and create more moments to match in the estimation process. Because prices change infrequently and with no regularity, I argue it is unlikely that consumers are considering cross-week substitution.

I include different vectors of station characteristics for $\theta_{\alpha_0,\beta_0}$ and Σ. I estimate homogeneous preferences for weekly price, reliability (station health index score over the prior four weeks), whether the station has a low, medium, or high health index score during the current week,[29] and station-hour fixed effects. The vector of characteristics estimated with heterogeneous preferences includes time of day (morning: 6-9 AM, late morning: 9 AM - 12 PM, afternoon: 12-4 PM, early evening: 4-7 PM, evening: 7-11 PM, and overnight: 11 PM - 6 AM) and charging network.

6.2.1 Instruments

I build a vector of instruments that incorporates both cost and demand shifters. Price is endogenous to demand, so higher priced products likely also have higher market shares. Regressing market shares on prices without an instrument would suggest that demand is increasing in price, even though we expect that it is likely decreasing in the price of a specific good. Instruments allow the econometrician to separate the endogenous and exogenous portions of price.

The goal of cost instruments is to capture the effect of marginal cost – rather than demand – on prices. I use the marginal cost of electricity that each station faces as an instrumental variable. This fully captures changes in marginal operating costs. There is variation in the marginal cost of electricity across hours and weeks, but there is very little within-week-hour across-station variation.

I rely on demand shifters to capture variation in exogenous price pressure across stations. My first set of demand instruments is a set of differentiation instruments, a geographical analog to those proposed by Gandhi and Houde (2020). I instrument for the number of own-network and competitor stations within 5-, 10-, 15-, and 20-minute rings of the station. These instruments are intended to account for the amount of differentiation and therefore price setting power a station has in the market. The final set of demand instruments is

[29] I use the categorical variable in an attempt to control for the share of chargers that were operable during the week rather than the share of charging attempts that were successful. Both are included in the calculation of the health index, and the effect of a broken charger on the index should be larger than the effect of a failed charging attempt. Drivers can learn of broken chargers from mobile applications and choose to avoid a station whereas they will suffer failures to charge at the station and may choose to use another charger at the station. I observe clumping just below health index scores of 50 and 75 in the data, which I believe is indicative that there are a large number of cases where one or two of four chargers are broken for the entire week and there is some variation in the number of failed charges that pushes the scores just below the corresponding share of chargers that are functional (for a station with three functioning chargers and one broken charger, this would be 75% which I expect maps to a health index score of 75).

Table 3: Estimated Demand System Parameters

Homogeneous Parameters (α, β)	Estimate	(SE)
Price	-2.655^{**}	(0.648)
Reliability	0.007^{**}	(0.001)
Health Index < 50	-0.447^{**}	(0.041)
Health Index 50-75	-0.172^{**}	(0.016)
Heterogeneous Parameters (σ)		
Morning	-0.117	(56.307)
Late Morning	0.318	(129.797)
Afternoon	-0.930	(66.499)
Early Evening	0.451	(21.700)
Evening	-0.175	(36.180)
Overnight	7.671	(4.323)
Electrify America	0.338	(10.612)
Tesla	0.041	(8.970)

Notes: $*/**$ Denote significance at $p < 0.05/0.01$ level.

the price of residential charging at different times of the day to capture changes in relative preference for the outside option.

6.3 Results

I present the estimated parameters of the model of consumer demand in Table 3. The homogeneous preferences (α, β) are calculated as the average preference across individuals. The average preference for reliability is positive and significant while the coefficients for stations with low or medium health scores are negative and significant. These results show that consumers value station functionality, which matches what one would expect. The average preference for price is negative and significant, suggesting general preferences for cheaper charging. These preferences are relatively small though. The implied willingness to pay for a 10 point increase in reliability (measured 0 to 100) is 2.6 cents and 16.8 cents for increasing the current week health index from below 50 to above 75.

Heterogeneous parameter estimates represent the standard deviation of preferences across individuals. The estimates for time of day all have large standard errors, which means I cannot rule out a large amount of heterogeneity in preferences for charging at different times of day. The results for network preferences are similarly inconclusive. One would expect that all or most of the σ parameters should be large. The results in Section 5.2 suggest strong network preferences and we would expect consumers to potentially have rigid preferences for

time of day as well. One possibility is that the lack of microdata or demographic data for fast charging consumers meaningfully reduces the precision of estimation. Another is that my data are noisy and also cannot capture the true demand at an hour where consumption is capacity constrained.[30]

I aggregate individual preferences to calculate elasticities for each individual and then average elasticities across individual. The model suggests that consumers have relatively homogeneous price preferences, and are somewhat price sensitive. I present average own- and cross-price elasticities in Table 4. Average own-price elasticities vary across networks and time of day, with estimates ranging from -0.99 to -1.78 and an average elasticity of -1.33.[31] The confidence intervals for these estimates are fairly large; the average elasticity has a 95% confidence interval of -0.66 to -1.87 (I provide confidence intervals for all own-price elasticity estimates in Table 6). These estimates are in line with the -1 to -1.45 Bernard et al. (2025) find in the United Kingdom.

I estimate that cross-price elasticities are very low, suggesting little willingness to substitute across stations or hours. For example, consider the case where a Tesla station raises the 12 PM price by 1% but does not change prices from 1-4 PM. The own-price elasticy suggests charging from 12-1 PM will decrease by 1.6%, but charging will only increase during 1-4 PM hours by an average of 0.007%. The cross-elasticities for within-network substitution and within-time of day substitution are also low. This results in a large amount of diversion to the outside good, which is likely charging at a home or slower public charger. Taken together, the own- and cross-price elasticities suggest consumers are somewhat price sensitive, but do not appear to search across stations for lower prices.

Elasticities vary across networks and time of day. The results suggest Tesla stations face more inelastic demand than EVgo and Electrify America stations. Overnight demand is more inelastic and substitution to other overnight hours is greater. This could reflect that charging during these times may disproportionately capture long distance driving, where the cost of delaying charging is greatest. Another explanation is that prices are lowest overnight, and

[30]The capacity constraint both censors and influences demand. I cannot observe individuals who would have charged if there had been more chargers, and some individuals may see that a station is busy and choose to go somewhere else. Conlon and Mortimer (2013) develops a method for handling these types of cases, but without data on when chargers were occupied, I cannot implement it. It is not clear which direction this unobserved demand will bias my results. The inclusion of station-hour fixed effects means that one might expect that a given station-hour is more likely to be capacity constrained when prices are lower, leading to estimating consumers are less elastic than they truly are. On the other hand, Figure 11 suggests networks may use prices as a rationing method meaning that demand would be underestimated at higher prices, leading to estimating that consumers are more elastic than they truly are.

[31]I do not focus on differences between subscriber and non-subscriber prices because the difference is a constant multiplier, so switching from non-subscriber prices to subscriber prices would hold all percentage changes in prices constant.

Table 4: Average Own- and Cross-Elasticities

Network	Time	Own	Within Station-Time	Within Network	Within Time
EVgo	Morning	-1.5268	0.0006	0.0007	0.0013
EVgo	Late Morning	-1.5922	0.0012	0.0007	0.0029
EVgo	Afternoon	-1.7087	0.0019	0.0007	0.0053
EVgo	Early Evening	-1.7800	0.0014	0.0007	0.0039
EVgo	Evening	-1.6618	0.0008	0.0007	0.0020
EVgo	Overnight	-1.3681	0.0042	0.0013	0.0080
Electrify America	Morning	-1.6080	0.0015	0.0019	0.0013
Electrify America	Late Morning	-1.6062	0.0033	0.0019	0.0030
Electrify America	Afternoon	-1.6038	0.0057	0.0021	0.0056
Electrify America	Early Evening	-1.6055	0.0040	0.0019	0.0040
Electrify America	Evening	-1.6074	0.0021	0.0019	0.0021
Electrify America	Overnight	-1.6014	0.0081	0.0028	0.0084
Tesla	Morning	-1.0553	0.0015	0.0023	0.0013
Tesla	Late Morning	-1.1692	0.0036	0.0023	0.0029
Tesla	Afternoon	-1.1815	0.0070	0.0025	0.0055
Tesla	Early Evening	-1.1838	0.0050	0.0023	0.0039
Tesla	Evening	-1.1439	0.0026	0.0023	0.0020
Tesla	Overnight	-0.9887	0.0100	0.0035	0.0083

the consumers who choose to charge at the cheapest and most inconvenient hours may have worse outside options, reducing their willingness to substitute away as prices increase (but remain the lowest of any hour). The latter explanation would suggest a form of nonlinearity in elasticities rather than a true preference for overnight charging.

7 Pricing Strategies and Efficiency

Descriptive evidence in Figure 1 shows large markups above marginal operating costs, and the empirical results in Section 5.2 suggest increasing the number of stations in the market does not decrease markups. It is important then to understand how efficient current prices are given they are unlikely to decrease as more stations enter. Using the results from Section 6.3, I calculate the most efficient prices that recover fixed costs and determine how far observed prices are from those that minimize deadweight loss.

7.1 Modified Ramsey Problem

The standard approach to calculating Ramsey prices, the set of linear prices that minimizes deadweight loss while recovering fixed costs, assumes quantity can be increased at a

finite marginal cost. To account for capacity constraints, I modify the Ramsey problem formulation from Crew, Fernando and Kleindorfer (1995) to include a condition that quantity demanded must be at or below the station's capacity, $\bar{X}$. I simplify this problem by writing it for a single charging station, but the network's problem is analogous.

During a day, a station supplies charging at a set of times (i.e. morning, afternoon, evening) $t \in 1, 2, ..., T$ with constant marginal cost for each given by c_t. Heterogeneous consumers each have demand for charging at different time periods, $\boldsymbol{x} = (x_1, ..., x_T)$, based on their individual valuation of consumption bundles, $V(\boldsymbol{x}, i)$, where i indexes consumers. The total demand for charging at time period t is the sum of individual demand, $x_t(\boldsymbol{P}, i)$, which is a function of all time period prices and individual preferences:

$$X_t(\boldsymbol{P}) = \int_I x_t(\boldsymbol{P}, i)di, \quad t \in 1, ..., T. \tag{12}$$

The constrained Ramsey problem is to choose the set of prices, $\boldsymbol{P}$, that maximize social surplus, $W(P)$, subject to capacity constraints and recovering fixed costs. I write this as:

$$\begin{aligned} \max_{\boldsymbol{P}} W(\boldsymbol{P}) = & \int_I \Big[V(\boldsymbol{x}, i) - \sum_t P_t x_t(\boldsymbol{P}, i)\Big]di + \sum_t (P_t - c_t)X_t(\boldsymbol{P}) \\ \text{s.t.} \quad & \Pi(\boldsymbol{P}) = \sum_t (P_t - c_t)X_t(\boldsymbol{P}) \geq \Pi_0 \\ & 0 \leq X_t(\boldsymbol{P}) \leq \bar{X}, \quad \forall t \in 1, ..., T \end{aligned} \tag{13}$$

where the two terms added together on the first line represent consumer surplus and station surplus, respectively. The two constraint conditions require that the station earn surplus to cover fixed costs and that the amount of charging cannot exceed capacity at any hour. Converting this maximization problem from an individual station to the network's price setting decision requires optimizing across all stations and times of day (t indexes station-times rather than just times at a single station) but otherwise does not change the problem.

Removing the profit restriction makes the problem analogous to the two-period problem depicted in Figure 5, where $\mathrm{P}^*_{\mathrm{P}}$ and $\mathrm{P}^*_{\text{O-P}}$ are efficient. Alternatively, in the absence of the capacity constraint, the problem reduces to the unconstrained (traditional) Ramsey problem. This means that if the quantities demanded at the unconstrained Ramsey prices are below the capacity constraint, the unconstrained Ramsey prices are the social surplus maximizing linear prices. However, in the case where the quantity demanded at period t at the unconstrained Ramsey prices, $\boldsymbol{P}^R$, exceeds capacity, the station needs to raise the price

so the market will clear. Doing so has the added benefit of generating revenue (without increasing deadweight loss), and lowers the optimal markup over marginal operating cost on other time periods relative to the unconstrained Ramsey price.

There are three possible forms for the solution to Equation (13). First, there is the case where the capacity constraint binds but the profit constraint does not. In all periods where demand intersects marginal cost on the vertical portion ($X_t(\boldsymbol{P}^R) > \bar{X}$), the station generates surplus of $(P_t - c_t)\bar{X}$, which sum to a value greater than Π_0. By raising prices to eliminate deadweight loss from misallocation, the network generates sufficient surplus to recover fixed costs and does not need to raise prices above the efficient prices. Second, if the efficient prices do not generate enough surplus and the capacity constraints do not bind on the unconstrained Ramsey prices, $\boldsymbol{P}^R$, then $\boldsymbol{P}^* = \boldsymbol{P}^R$. Third, if both constraints bind, then the solution is to use the elasticities of demand as in unconstrained Ramsey pricing to raise prices beyond the market clearing prices and generate the amount of surplus still required to meet the profit target. In this case, setting prices to allocate charging efficiently recovers some, but not all, of the fixed costs. The remaining fixed costs are recovered in the way that generates the least deadweight loss following the Ramsey principle of larger markups on more inelastic demand.

A limitation of my data is that there is no obvious way to define $\bar{X}$. In Section 5.2.5, I defined a station's maximum feasible draw as the 98th percentile of hourly daytime electricity consumption. Setting $\bar{X}$ at the maximum feasible draw multiplied by the number of hours (i.e. four afternoon hours per day times seven days per week) would require the station to operate with all chargers in use and little customer idling for a whole week. If there is a large amount of stochastic variation in demand, car types, or idling, then it's possible for the station to be well below that level and still run out of capacity regularly.[32] I therefore set $\bar{X}$ as 70% of maximum capacity over a period to account for variation in consumption. I discuss the implications of this choice when I present results in Section 7.2.

7.2 Comparison of Observed and Efficient Prices

I use the elasticities of demand and observed prices, costs, and quantities to solve for profit maximizing, social surplus maximizing, and constrained Ramsey prices. I assume linear demand and calculate profit maximizing prices as the set of prices that maximizes the profit portion of Equation (13) subject to the capacity constraint. These are static profit maximizing prices that would only maximize profit if competition is not dynamic. I argue this is a valid approach because the results in Section 5.2 suggest stations do not compete

[32]It seems likely there is significant stochastic variation. Tesla has switched from time-of-use pricing to demand-based pricing, which would not be necessary if there was no stochastic variation in demand.

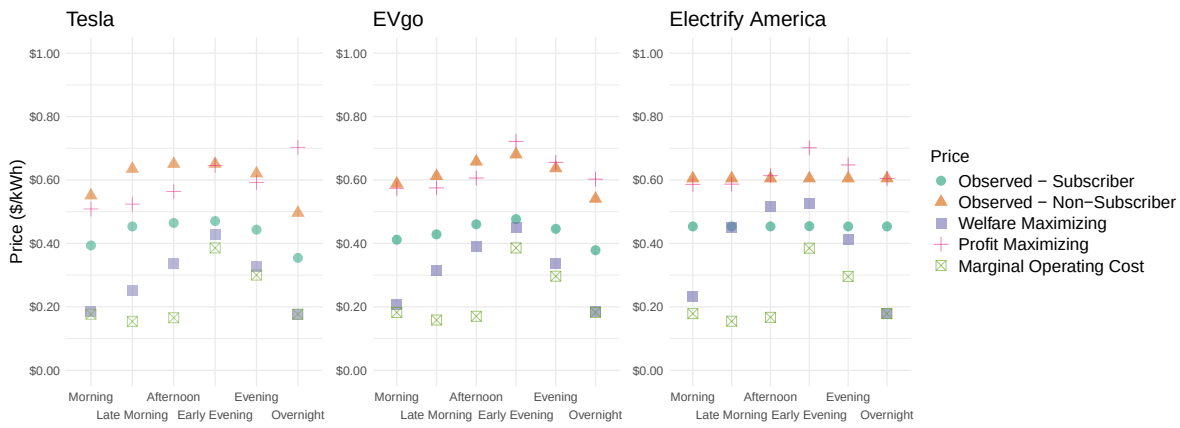


Figure 12: Comparison of Observed and Profit and Social Surplus Maximizing Prices

on prices and Figure 2 shows networks do not follow each other's price changes.[33] I calculate social surplus maximizing prices as the set of prices that maximizes Equation (13) subject to only the capacity constraint. The constrained Ramsey prices are the set of prices that maximizes Equation (13) subject to both the capacity constraint and restriction that profits exceed some level in order to recover fixed costs.

I start by comparing observed prices to both the set of prices that maximize profit and the set that maximize social surplus. In Figure 12, I plot the average prices and marginal cost by network and time of day, using the same divisions as in the demand estimation. I show both the average observed price that a network's subscriber would pay and that a driver without a subscription would pay. Across all networks and times of day, the average observed non-subscriber price is similar to the average profit maximizing price.[34]

The alignment of observed non-subscriber prices and estimated profit maximizing prices provides a measure of validation for the demand estimation results.[35] Networks that are vertically integrated may want to increase car sales by lowering charging prices to increase the value of owning their car. However, networks tend to provide at least two years of free network subscription (access to the subscriber prices), so they would likely use the subscriber price rather than non-subscriber price to increase the consumer's value of the vehicle. Therefore, setting non-subscriber prices at a level that maximizes the network's profit in the charging market is aligned with the network's overall profit maximizing objective.

[33]The results in both Section 5.2 and Section 6.3 show consumers substiute very little across networks, so making the model dynamic would have little downward pressure on prices.

[34]The one exception to this is overnight at Tesla stations, when demand is more inelastic and prices are low. It is possible prices are lower to encourage consumers to switch from peak-demand hours to overnight hours or to avoid displeasure from consumers on roadtrips who need to charge overnight. Given that few drivers charge overnight, there is relatively little revenue to be made from raising prices during those hours.

[35]My demand estimation model does not have a supply side, so there is no assumption that the prices are profit maximizing.

The difference between marginal operating costs and welfare maximizing prices reflects capacity constraints.[36] If a station is not at capacity at a given time, then the social surplus maximizing price is equal to the marginal operating cost. When the station faces capacity constraints, marginal cost becomes vertical and the social surplus maximizing price is the price that equates quantity demanded and maximum capacity. The more that capacity constraints bind during a time of day, the higher efficient prices will be at that time.

A second result evident in Figure 12 is that subscriber prices are generally above the welfare maximizing prices. Additionally, there is much more within-day variation in welfare maximizing price than in observed prices. Network behavior is not consistent with using two-part tariff pricing. An important comparison then is between observed prices and the linear prices that produce the greatest social surplus while still recovering a station's fixed costs, the Ramsey prices. If prices are notably above the Ramsey prices, then there is room to increase social surplus by decreasing prices.

I do not observe the cost of building or maintaining stations, so I consider a range of fixed cost assumptions. Tesla claimed in 2025 that their average cost to build a station was about \$40,000-45,000 per charger (Miyatra, 2025), which I use as a low-end construction cost estimate because it is lower than academic estimates (Gamage, Tal and Jenn, 2023; Li, 2023). As the upper end, I use the 75th percentile of National Electric Vehicles Infrastructure (NEVI) site cost per charger, which is \$233,000 (McDonald, 2024). I use the 75th percentile because NEVI funding was limited to stations on designated alternative fuels corridors, which are likely to require more grid upgrade and construction costs than the stations in my data.

The AFDC estimates maintenance costs at about \$400 per charger month (AFDC, 2023) and I assume that stations pay monthly capacity charges for 200 kW per charger. Using a frequently assumed lifespan of 10 years and discount rate of 6%, I calculate the net-present value of lifetime costs as \$109,000 under the low-end estimate and \$302,000 under the upper-end estimate. For reference, Li (2023) and Gamage, Tal and Jenn (2023) estimate per charger costs at \$152,000-302,000 and \$122,000-440,000, respectively. My fixed cost estimates are slightly lower, which partially reflects that the stations in my sample are notably larger (9.9 chargers per station) than those in Li (2023) (1.7 chargers per station) and Gamage, Tal and Jenn (2023) (1.5 chargers per station). I translate my bounds to weekly costs and set the amount of fixed-cost recovery for the low, medium, and high fixed cost scenarios at \$210, \$395, and \$580 per charger per week, respectively.

Capacity constraints put upward pressure on afternoon and evening prices. In Figure 13, I plot the welfare maximizing linear prices that recover fixed costs under a medium fixed

[36] I show in Figure 32, that changing the capacity threshold from 70% to 60% or 80% of feasible maximum charging changes welfare maximizing prices by a few cents.

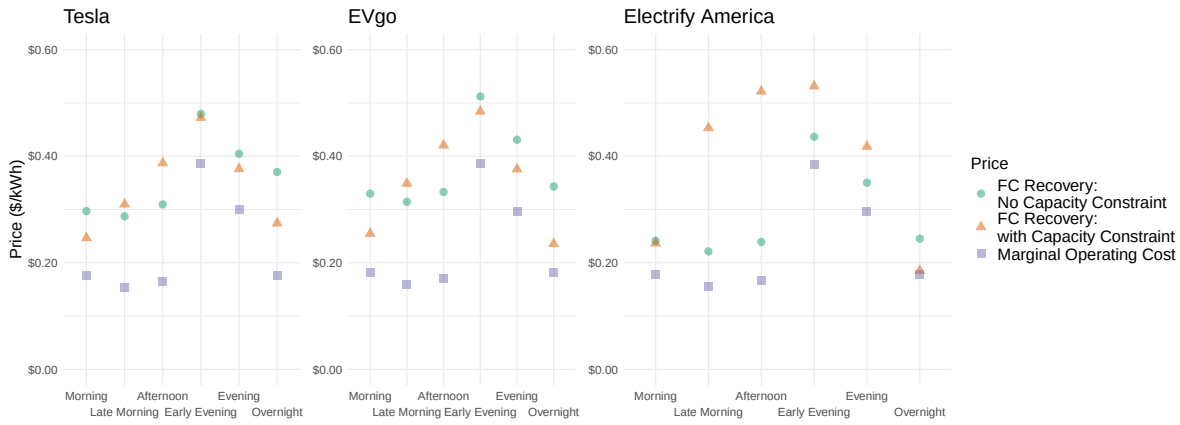


Figure 13: Comparison of Welfare Maximizing Prices, with and without Capacity Constraints

cost assumption. In the absence of capacity constraints, variation in Ramsey price markups reflects only variation in the elasticity of demand, so the markups would be fairly consistent throughout the day. However, incorporating capacity constraints raises Ramsey prices during congested times to ensure market clearing and reduces them during less busy times.

There is room for networks to lower non-subscriber prices while still recovering fixed costs. Figure 14 shows average observed prices and constrained Ramsey prices under different fixed cost assumptions. The difference between observed and Ramsey prices grows nonlinearly as fixed costs increase. At low fixed costs, nearly all stations can recover prices with market clearing prices. As assumed fixed costs increase, a greater share of stations must apply additional markups to recover fixed costs. Under the medium fixed cost assumption, Ramsey prices are generally similar to efficient prices and at least 10 cents below non-subscriber prices.

It is worth noting that not all stations successfully would successfully recover fixed costs under these scenarios. For example, Figure 33 shows that while all Electrify America and the vast majority of Tesla stations recover fixed costs across all three scenarios, only half of EVgo stations would be able to recover high fixed costs. The difference between EVgo and the other two networks may reflect that their stations often use lower powered chargers (which are cheaper to build and require lower monthly capacity charges) and are more likely to receive state or federal subsidies. Subsidy increase the probability a station recovers fixed costs and shift the Ramsey prices down because there are lower fixed costs to recover. Using lower powered chargers and receiving subsidies could even reduce the network's true level of fixed costs the below those in the low fixed cost scenario.

I plug the Ramsey prices back into the demand estimation model to calculate the effects on surplus. I estimate that charging the Ramsey prices instead of observed prices would increase social surplus by 7-10% depending on the fixed cost assumption. The increase to consumer surplus is much larger (31-72%), with slightly less than half coming from recovered

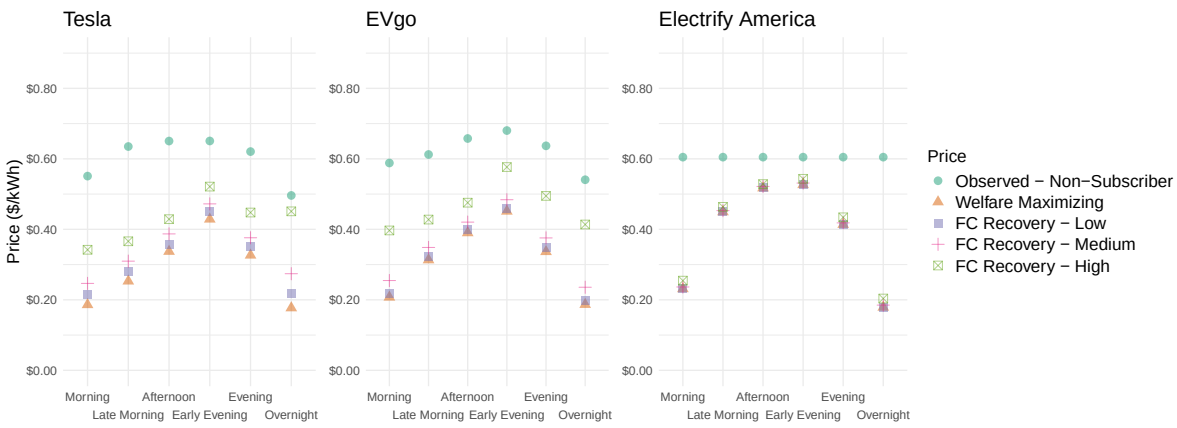


Figure 14: Comparison of Observed and Ramsey Prices

deadweight loss. The remaining increase is a large transfer of surplus from producers, who lose 9 to 32% of observed surplus. This transfer would reduce networks' incentives to build stations and could hinder EV adoption targets.

No network appears to follow a two-part tariff or vertical integration strategy. Two-part tariffs are more efficient than linear prices for fixed-cost recovery and often notably reduce deadweight loss. Figure 14 shows that under low and medium fixed cost assumptions, the optimal linear prices are almost equal to the efficient prices. Two-part tariffs provide a minimal social surplus improvement over Ramsey prices in those cases. This suggests that while capacity constraints bind, linear pricing could achieve nearly all of the social surplus benefits of two-part tariffs pricing.

8 Conclusion

Standard microeconomic models would suggest increasing the number of competitors in the EV fast charging market would drive down prices, but prices have remained high as stations have entered. I show that this is because consumers and stations do not respond to the entry of new stations operated by competing networks. I estimate that quantities decrease at incumbents following own-network entry. Drivers only respond to new stations from the network whose stations they already use, suggesting they do not substitute across networks. I show that this result is not due to differences in charging standards across stations but rather reflects network preferences. There are a number of possible reasons for low cross-network substitution: most networks only post prices in their own mobile application, subscribers may have a subscription to one network, and some some cars came with some amount of free charging at specific networks, which could lead to inelastic demand.

Given the lack of price competition and prevailing high markups above marginal operating

costs, I investigate how far prices are from surplus maximizing prices. I estimate a model of demand to recover own- and cross-price elasticities. I find own-price elasticities ranging from -0.99 to -1.78 across networks and times of day, consistent with the -1 to -1.45 Bernard et al. (2025) estimate. I embed the elasticities into a welfare maximization problem to determine the sets of profit maximizing, surplus maximizing, and Ramsey prices.

Observed prices are very similar to the set of linear prices that would maximize profits. There is no evidence that networks are using a two-part tariff system or vertical integration to maximize profits. The choice to use linear prices creates deadweight loss, but I show that at low fixed costs, the deadweight loss is minimal because the most efficient linear prices and two-part tariff prices nearly coincide. This is because capacity constraints push efficient prices above marginal operating costs to ensure markets clear. As fixed costs increase, the deadweight loss from use of linear prices increases. I estimate that switching to Ramsey prices would increase social surplus by 7-10% depending on the level of fixed costs.

The paper's title asks what is driving high EV fast charging prices? The quantity-weighted average markup above marginal operating cost is 109%. A 48% markup is needed to clear markets and eliminate deadweight loss from misallocation. Beyond that, recovering fixed costs with linear prices requires a 53-98% markup depending on the fixed cost assumptions. Capacity constraints and fixed-cost recovery only explain about one third and one to two thirds of observed markups, respectively. The answer then appears to be that a lack of competition due to rigid consumer preferences is driving high markups.

The most common policy tool to address high prices is increasing competition. However, the entry analysis revealed that market entry does not increase competition between stations. Focusing on the lack of substitution may provide a direction for policy makers worried about high prices. It is possible that the high search costs for finding stations and prices may deter substitution. Regulation that requires stations to make prices accessible through an API so that third-party applications can post prices for all stations could increase competition. Any policies that successfully lower prices could lead to decreased station entry, although reducing search costs could also reduce barriers to entry for smaller networks.

My results suggest that a large share of markups are efficient due to insufficient capacity during peak demand periods. Policy focusing on fast charging should not only address prices but also the net supply of charging. Governments may want to offset potential losses due to pro-competition policies driving down prices by providing larger entry subsidies. Incentivizing more utilization of charging stations during off-peak hours is also a promising option. Increasing data sharing requirements could reduce search costs, which might increase cross-station and cross-hour substitution, potentially reducing the need for more chargers and increasing station surplus, thus reducing the need for subsidies.

Data on the fast charging market is sparse. In this paper, I am able to delve into prices in the fast charging market due to a large data collection effort. However, more data is needed on the costs that stations face and the decisions consumers make to understand the value of government investment in fast charging infrastructure and assess charging station policies. For example, consumer choice data would allow researchers to develop demand models that paint a more complete picture of how drivers substitute between charging locations. Given that low substitution is driving the need for more fast charging infrastructure due to capacity constraints at peak-demand hours, this research could be highly impactful. Partnerships with charging networks or government agencies providing subsidies could be promising for future research on consumer behavior in the fast charging market.

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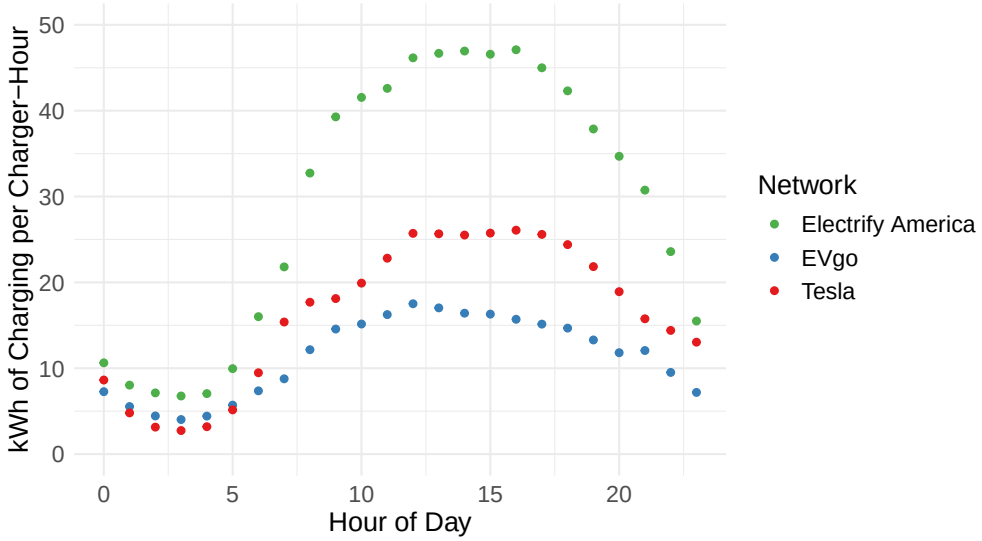
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A Appendix

Table 5: Charging Levels

Charger Level	Typical Power Output	Estimated BEV Charge Time	Estimated Electric Range per Hour of Charging	Typical Location
Level 1	1 kW	40-50 hours	2-5 miles	Home
Level 2	7-19 kW	4-10 hours	10-20 miles	Home, Workplace, and Public
DC Fast Charging	50-350 kW	20 - 60 min.	180-240 miles	Public

Notes: Source: U.S. Department of Transportation.
Charge time is calculated assuming the user charges a 60-kWh battery from 20-80% capacity. Most of the EV models popular in the U.S. have battery capacities around 60-80 kWh.



Notes: Data is for October 2025. Electricity utilization per charger is calaculated at each station and averaged across all stations operated by each network that are served by PG&E.

Figure 15: Average Hourly Electricity Use per Charger at DCFC Stations Served by PG&E

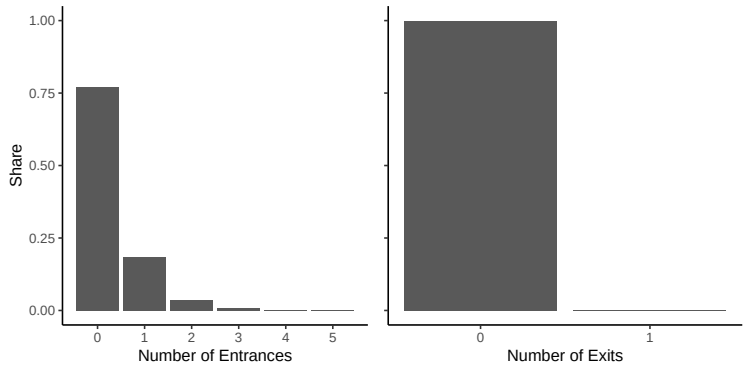
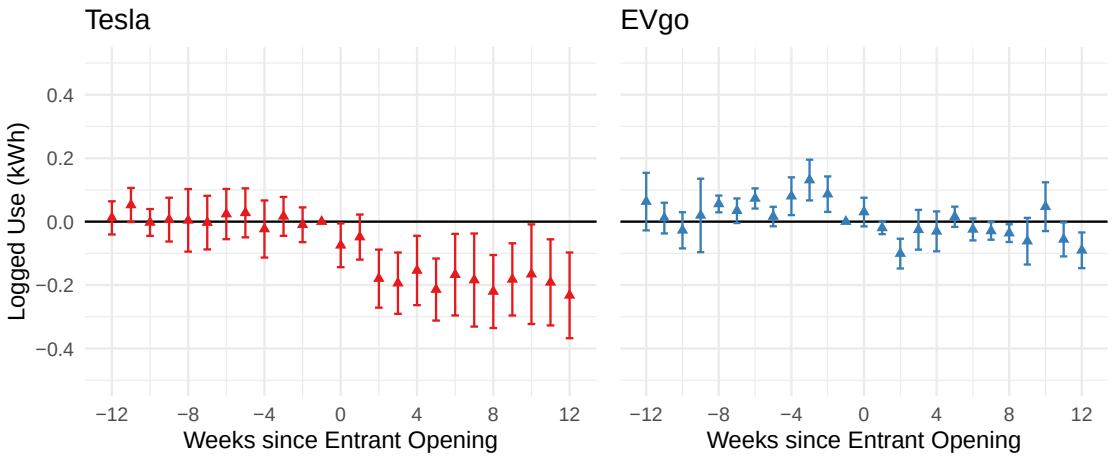
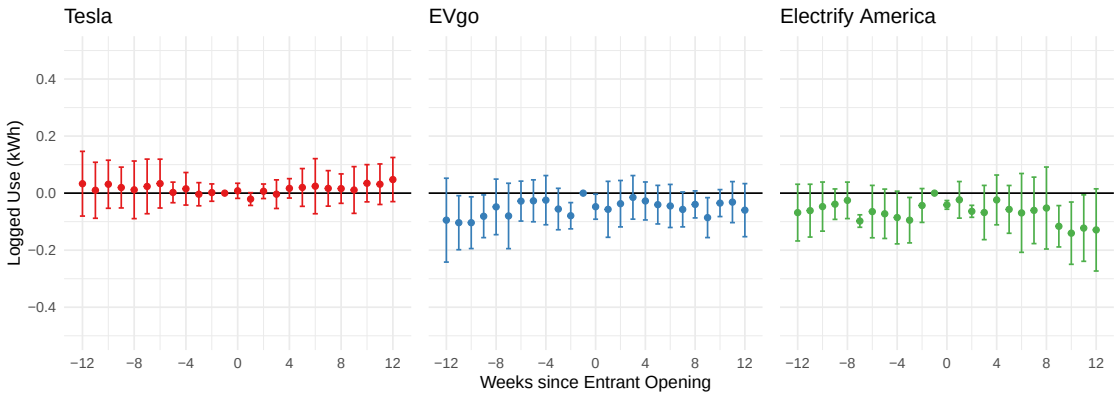


Figure 16: Frequency of Number of Entries and Exits Experienced by Incumbents



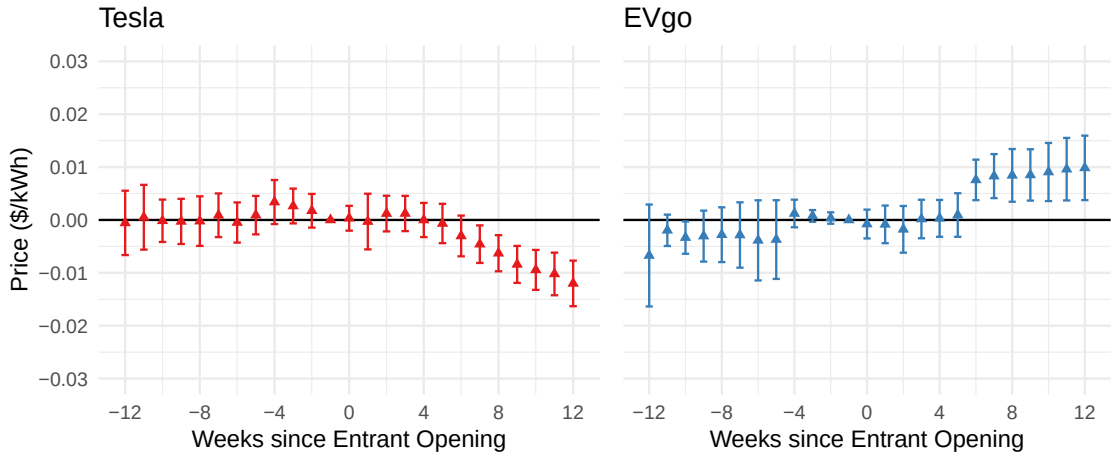
Notes: Data are total electricity consumption at each station and spans January 2022 - December 2025. Nearby entry is limited to stations within five minutes of an entrant. Electrify America does not open enough stations near other Electrify America stations to estimate an effect for the network.

Figure 17: Event Study Estimates of Own-Network Station Entry on Incumbent Hourly Electricity Consumption, by Network



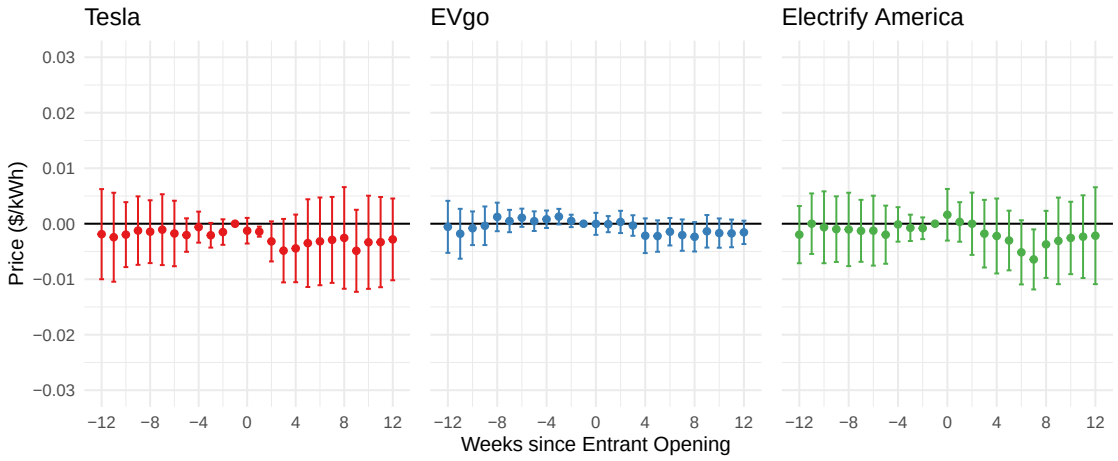
Notes: Data are total electricity consumption at each station and spans January 2022 - December 2025. Nearby entry is limited to stations within five minutes of an entrant.

Figure 18: Event Study Estimates of Competitor Station Entry on Incumbent Hourly Electricity Consumption, by Network



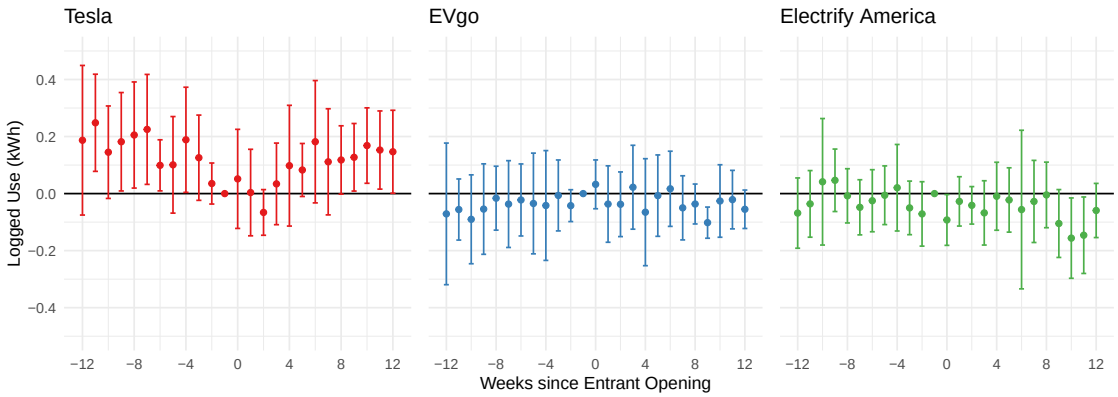
Notes: Data are station-week-hour prices and spans October 2024 - May 2026. Nearby entry is limited to stations within five minutes of an entrant. Electrify America does not open enough stations near other Electrify America stations to estimate an effect for the network.

Figure 19: Event Study Estimates of Own-Network Station Entry on Incumbent Hourly Prices, by Network



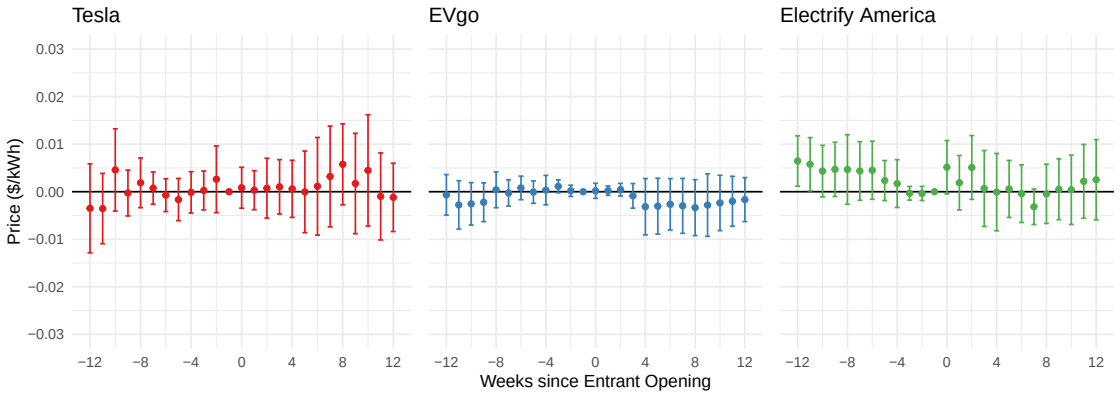
Notes: Data are station-week-hour prices and spans October 2024 - May 2026. Nearby entry is limited to stations within five minutes of an entrant.

Figure 20: Event Study Estimates of Competitor Station Entry on Incumbent Hourly Prices, by Network



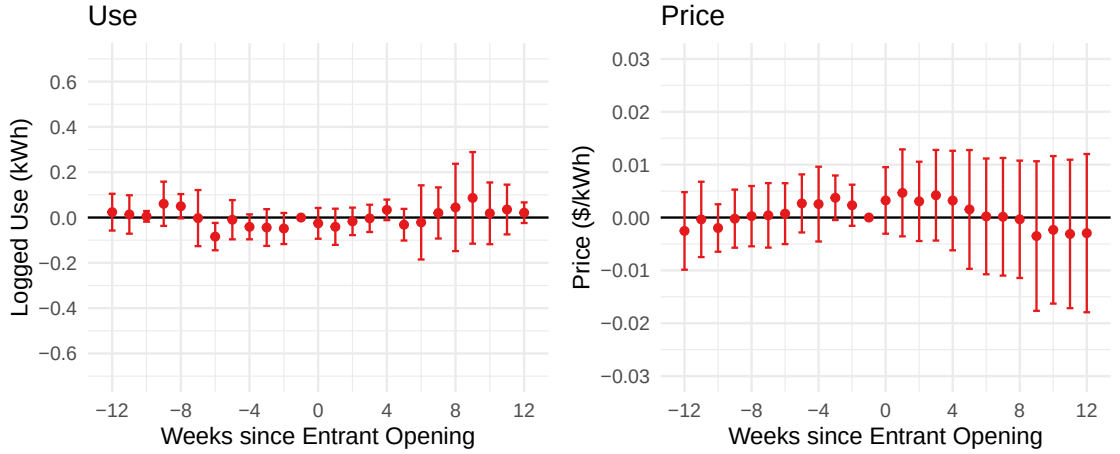
Notes: Data are total electricity consumption at each station and spans January 2022 - December 2025. Nearby entry is limited to stations within five minutes of an entrant.

Figure 21: Event Study Estimates of Entry of Competing Station with Same Charging Standard within Five Minutes on Hourly Electricity Consumption, by Network



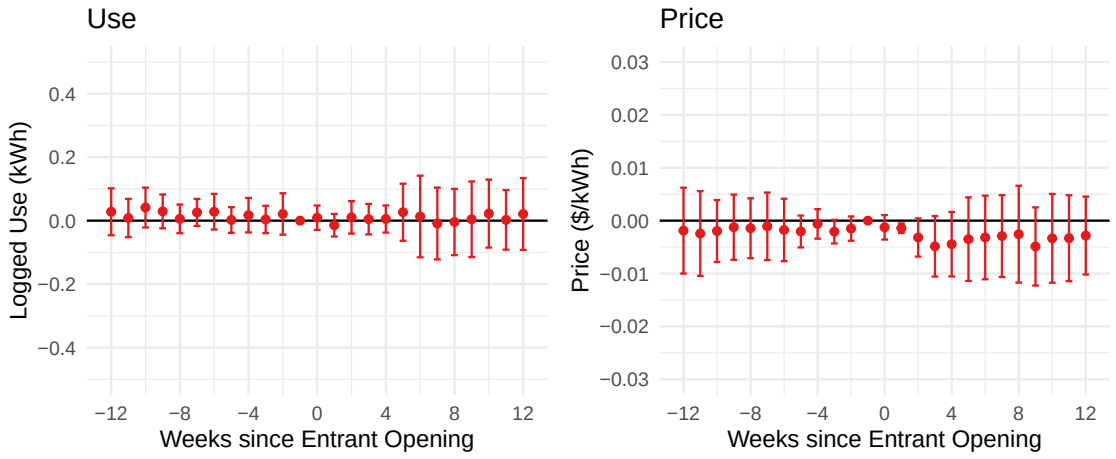
Notes: Data are station-week-hour prices and spans October 2024 - May 2026. Nearby entry is limited to stations within five minutes of an entrant.

Figure 22: Event Study Estimates of Entry of Competing Station with Same Charging Standard within Five Minutes on Hourly Prices, by Network



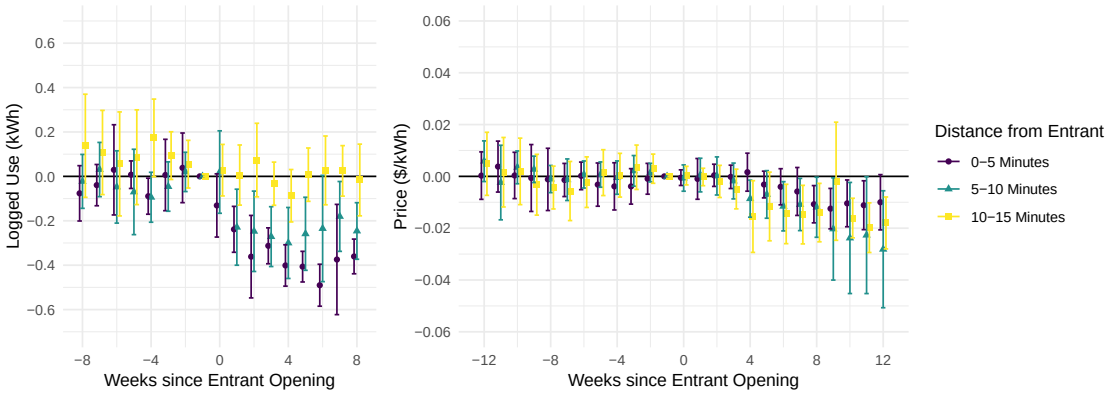
Notes: Data are hourly electricity consumption and prices at each station and span January - December 2025 and January 2025 - May 2026, respectively. Nearby entry is limited to stations within five minutes of an entrant.

Figure 23: Event Study Estimates of Charging Use and Price at Tesla Stations of Station Adding NACS Chargers



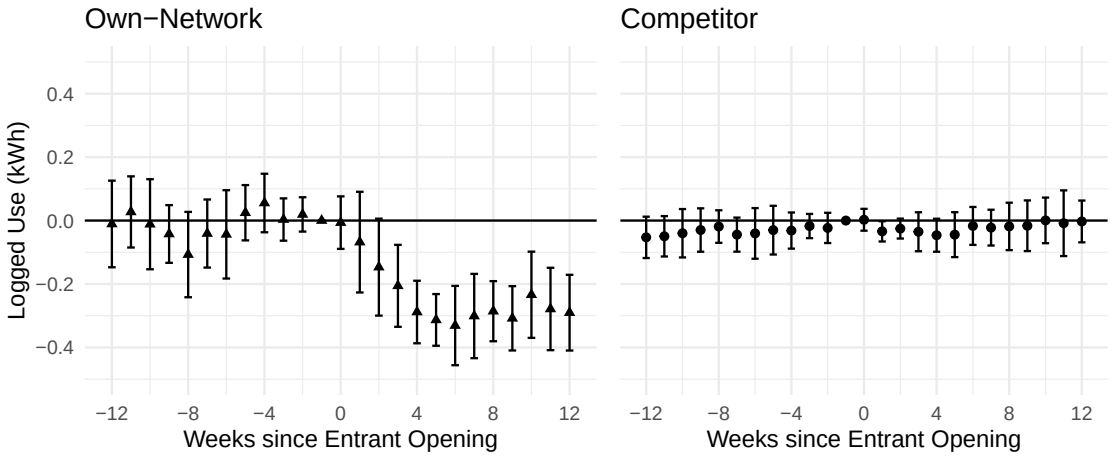
Notes: Data are hourly electricity consumption and prices at each station and span January - December 2025 and January 2025 - May 2026, respectively. Nearby entry is limited to stations within five minutes of an entrant.

Figure 24: Event Study Estimates of Charging Use and Price at Tesla Stations of Entrant in 2025 or Later



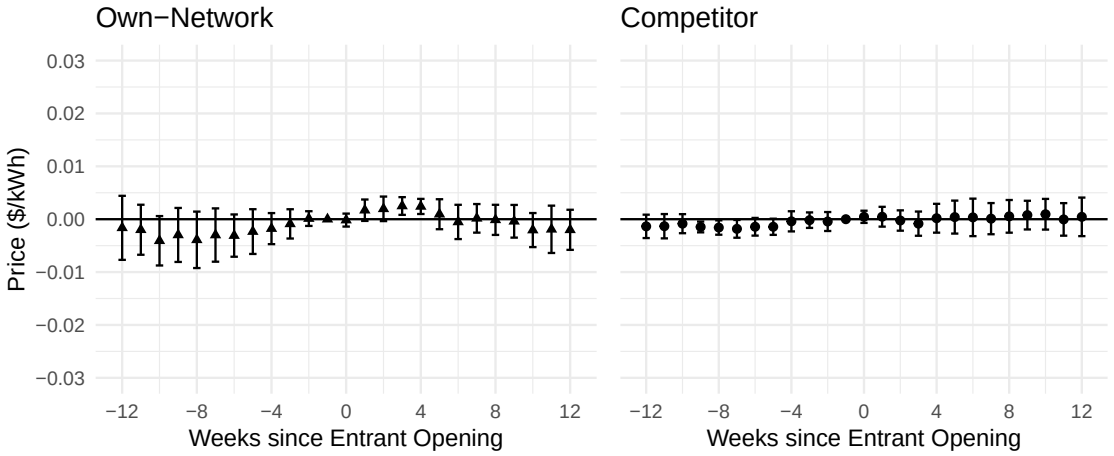
Notes: Data are hourly electricity consumption and prices at each station and span January - December 2025 and January 2025 - May 2026, respectively. Nearby entry is limited to stations within five minutes of an entrant.

Figure 25: Event Study Estimates of Tesla Own Station Entry within Five Minutes on Incumbent Hourly Electricity Consumption and Price by Distance



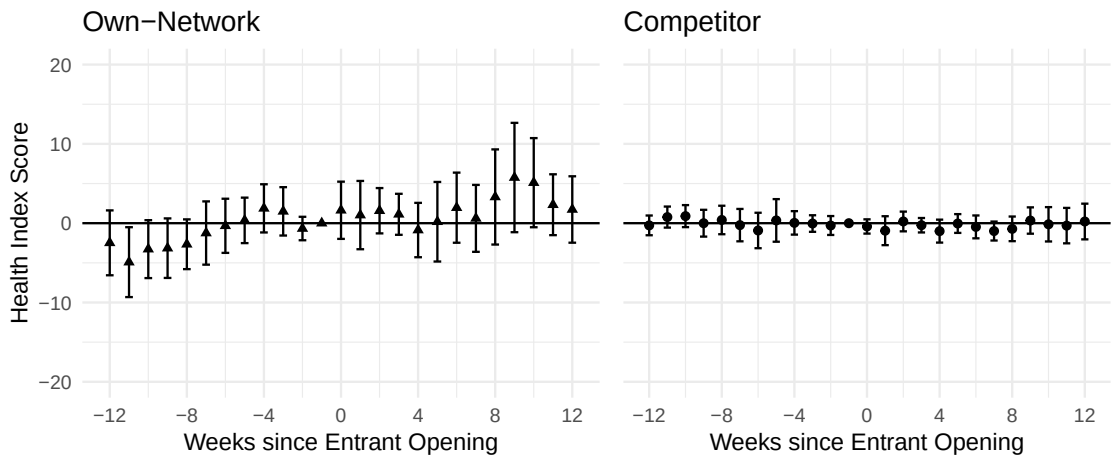
Notes: Data is for total electricity consumption at each station and spans January 2022 - December 2025.

Figure 26: Event Study Estimates of Own-Network Station Entry within Ten Minutes on Incumbent Hourly Electricity Consumption



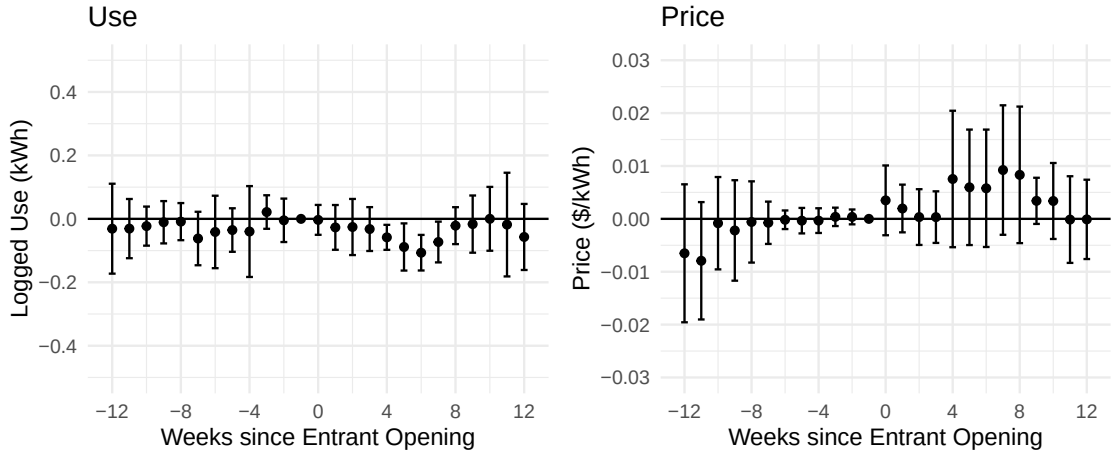
Notes: Data is for station-week-hour prices and spans October 2024 - May 2026.

Figure 27: Event Study Estimates of Own-Network Station Entry within Ten Minutes on Incumbent Price



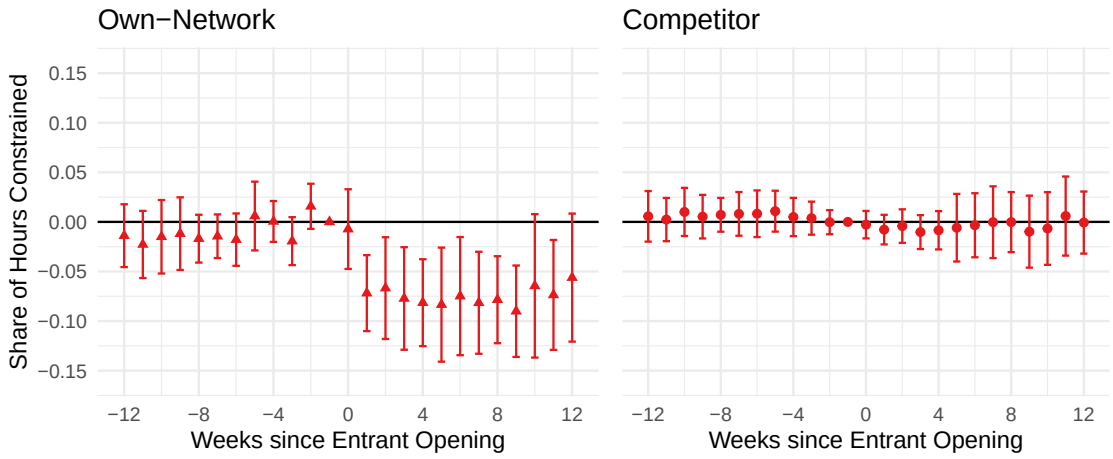
Notes: Data is for station-week health index score and spans October 2024 - May 2026.

Figure 28: Event Study Estimates of Station Entry within Ten Minutes on Station Quality



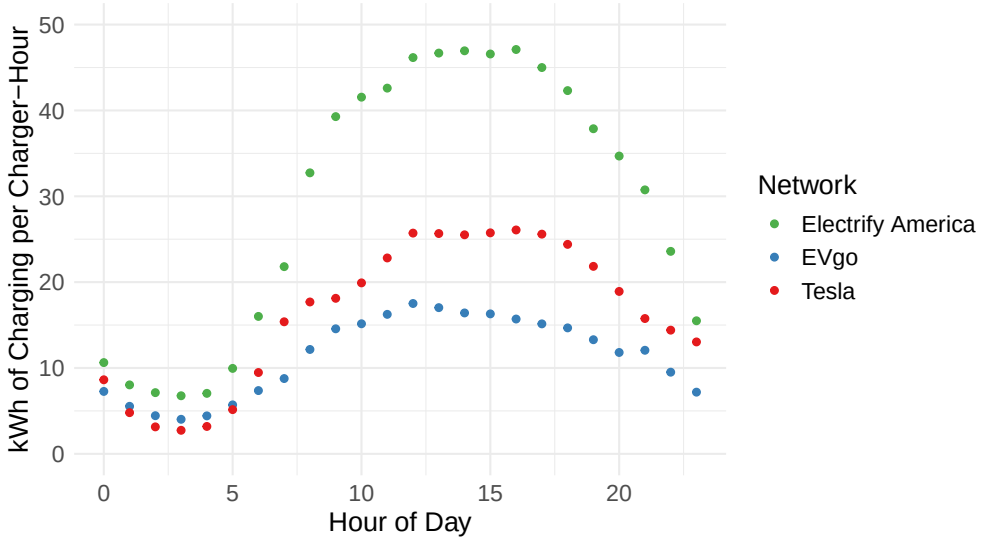
Notes: Data is for total hourly electricity consumption at each station and spans January 2022 - December 2025.

Figure 29: Event Study Estimates of Entry of Competing Station with Same Charging Standard within Ten Minutes



Notes: Data is for total hourly electricity consumption at each station and spans January 2022 - December 2025.

Figure 30: Event Study Estimates of Station Entry within Ten Minutes of Tesla Station on Share of Hours near Charging Capacity

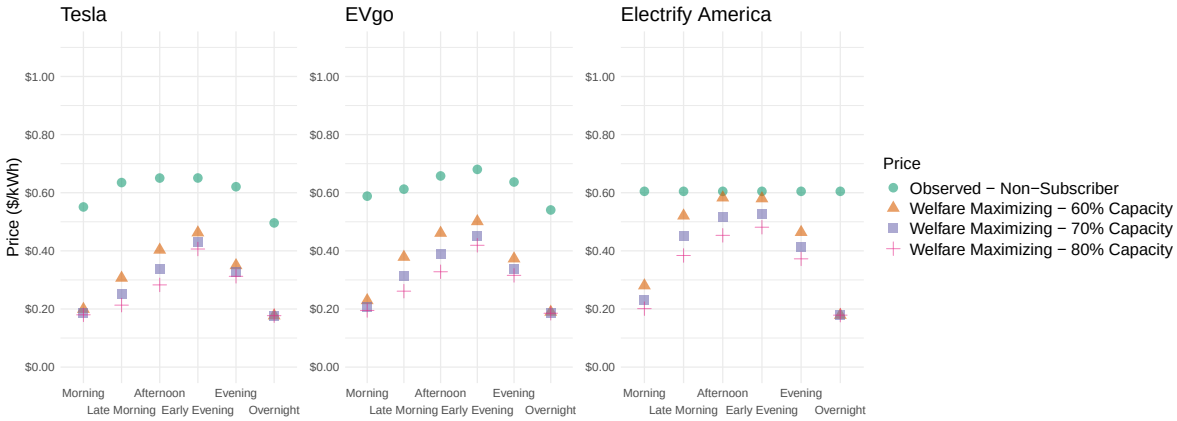


Notes: Data are for October 2025. Electricity utilization is the charger-weighted average across all stations operated by each network in PG&E's service area.

Figure 31: Average Hourly Electricity Use per Charger at DCFC Stations in PG&E Service Area

Table 6: Own-Price Elasticity Estimates and Confidence Intervals

Network	Time	Central Estimate	Confidence Interval
All	All	-1.334	(-0.660, -1.886)
EVgo	Morning	-1.527	(-0.722, -2.087)
EVgo	Late Morning	-1.592	(-0.749, -2.152)
EVgo	Afternoon	-1.709	(-0.830, -2.356)
EVgo	Early Evening	-1.780	(-0.852, -2.421)
EVgo	Evening	-1.662	(-0.806, -2.300)
EVgo	Overnight	-1.368	(-0.700, -1.991)
Electrify America	Morning	-1.608	(-0.748, -2.134)
Electrify America	Late Morning	-1.606	(-0.750, -2.134)
Electrify America	Afternoon	-1.604	(-0.767, -2.191)
Electrify America	Early Evening	-1.605	(-0.751, -2.138)
Electrify America	Evening	-1.607	(-0.770, -2.186)
Electrify America	Overnight	-1.601	(-0.801, -2.315)
Tesla	Morning	-1.055	(-0.523, -1.500)
Tesla	Late Morning	-1.169	(-0.580, -1.665)
Tesla	Afternoon	-1.182	(-0.594, -1.704)
Tesla	Early Evening	-1.184	(-0.591, -1.693)
Tesla	Evening	-1.144	(-0.576, -1.644)
Tesla	Overnight	-0.989	(-0.507, -1.455)



Notes: Capacity percentage is how the the percentage of maximum feasible charging used to define capacity. Setting capacity at 60% of maximum feasible charging makes the capacity constraight bind at lower quantities than setting capacity at 70 or 80%.

Figure 32: Welfare Maximizing Prices under Alternative Capacity Constraints

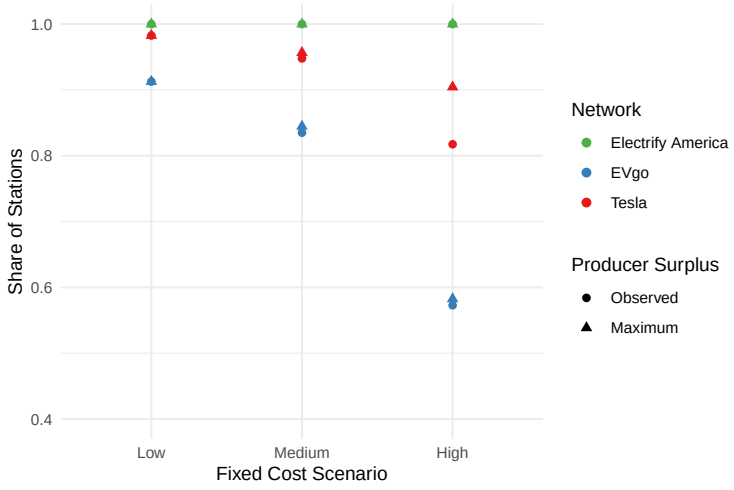


Figure 33: Share of Stations Able to Recover Fixed Costs

B Data Appendix

B.1 Price Data

Each week, a team of research assistants and I collected hourly prices for all Tesla, EVgo, and Electrify America DCFC stations in California. Data comes from each company's mobile application (Figure 34 provides a visual of how prices look in the applications) and PlugShare, a third party aggregator of charging station data, which posts EVgo prices. Each of these sources lists prices for each hour during the given day, which allowed collectors to pull all prices for a station at once.

The list of stations comes from querying the Alternative Fueling Station Location API from the DOE's Alternative Fuels Data Center (AFDC) each Saturday. Each data collector would check the hourly prices for a station from the list and enter all 24 prices into a spreadsheet for that week. Data collectors would repeat this process for all stations each week.

The collection period ran from Saturday to Friday each week. Each Saturday I would collect the list of stations from the AFDC, add them to a spreadsheet, and send out a list of collection assignments. Research asssitants were told the deadline for data collection was Thursday, and were in charge of collecting prices for around 100 stations between Sunday and Thursday (there were usually four students collecting EVgo prices and six students collecting Tesla prices, and I collected all Electrify America prices). Friday acted as a buffer day to allow me to collect any missing data and spot check the collected data.

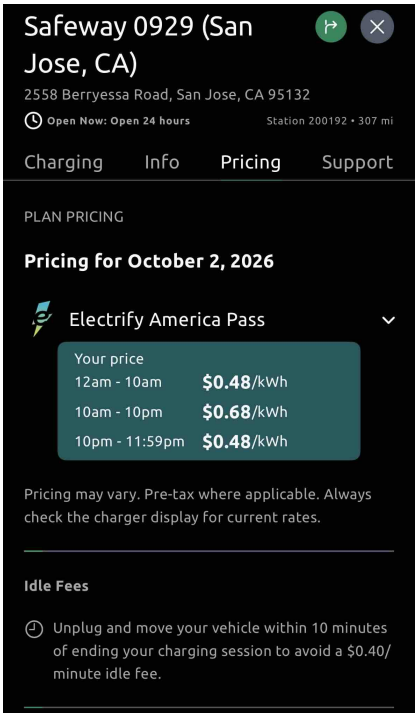
I piloted the data collection process from July through September of 2024 by collecting prices for Southern California stations. This helped create processes for checking data veracity. The data collection spreadsheet included a mechanism to compare against the prior week's prices to make error identification easier. However, there are likely still errors in the data. I compared my data against data from Paren for September of 2025 and found a greater than 99% match rate across all station-hours during the month, which suggests errors were very infrequent.

I have good coverage of stations – the data include prices for 1407 out of 1414 Electrify America, EVgo, and Tesla stations listed in the AFDC data at any point between October 2024 and May 2026. I do not have data on the other seven stations because they did not list prices on their network's application. This most likely means that the AFDC list includes some stations that are not operational and not that networks did not post prices for all stations. My final dataset cover 657 Tesla, 454 EVgo, and 296 Electrify America stations from October 2024 through May 2026, amounting to 2.6 million station-week-hour prices.[37]

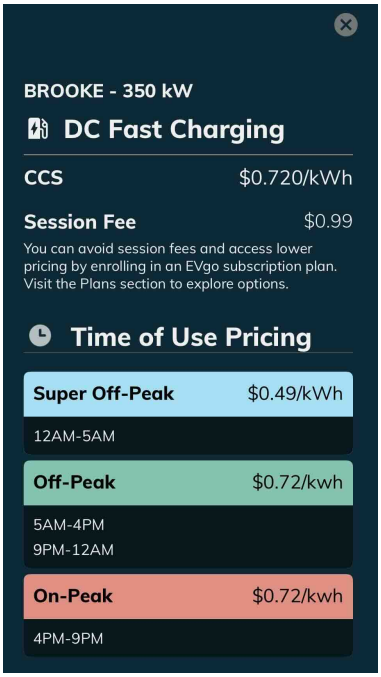
[37]We also collected prices for an additional 52 one-charger EVgo locations that I do not include in this

Prices could be collected any day between Saturday and Friday, and data collectors did not all choose to collect prices the same day. Therefore, it is important to be clear on interpretation of a week's prices as point in time prices collected during that week. My recorded weekly prices at a station will not capture price changes that occured that week if the changes occurred after data was collected. Further, it is possible that during a week where a network changed many stations' prices, some data collectors recorded new prices while some recorded old prices.

Panel A: Electrify America



Panel B: EVgo



Panel C: Tesla

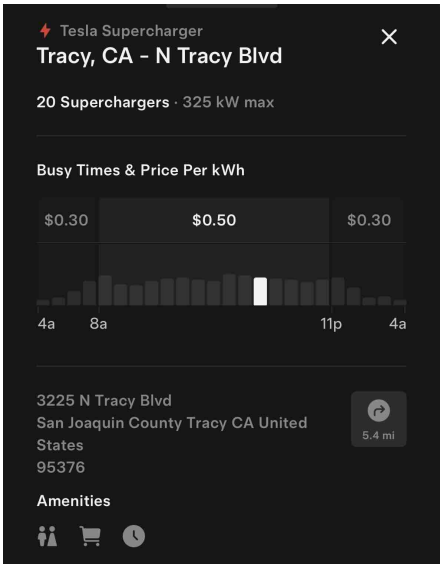


Figure 34: Price Presentation in Mobile Applications

B.2 Utility Data

Pacific Gas & Electric Company (PG&E) provided metered electricity use data for this project. They compiled all data for meters linked to the list of station addresses from the AFDC and all meters flagged in their data as providing public vehicle charging. Limiting to just stations in PG&E's territory, I am able to match 59% of stations from my price data (EVgo, Electrify America, and Tesla) and 43% of stations from the AFDC data (all stations).

analysis because I focus on locations with two or more chargers.

The low match rates reflect both that I cannot identify all stations in the PG&E data and that some stations may not be served by PG&E. It is possible that even stations PG&E serves will not be in the data they share with me because the station's meter is not at the address listed in the AFDC and the meter is not flagged as for a charging station. Because PG&E sent me data for all meters at the addresses of stations and many are located in large shopping developments, I have meters that are not used for stations. PG&E gives a 1-5 score for likelihood the meter is connected to a charging station. I only keep meters when it is clear that a station is the only customer connected to that meter. This means I am not able to identify many smaller stations that share meters with other businesses and some cases where multiple stations share a parking lot.

One additional caveat is that individual accounts may have multiple meters. Many stations have meters replaced at some point during the sample period, so I build a crosswalk of meters to stations. I aggregate meter data to the station level and use the AFDC station identifier to PG&E station identifier to merge price and use data.

I aggregate all data to the meter-hour level because some meters report hourly and some report every fifteen minutes. The electricity use is raw, and PG&E makes corrections to it before sending bills to customers. However, corrected billing volumes are only available aggregated to the billing period, so I use the raw volumes and acknowledge that they may introduce additional noise into the data.

To calculate hourly charging, I subtract out the baseline usage for each station. I calculate baseline hourly usage as the lowest use at a station. If a station has drivers charging for the entire sample period, I would overestimate the baseline, but this is highly unlikely as there is very little use overnight. If a station's lights are on the same meter as the chargers and are only on overnight, I will slightly underestimate the amount of overnight charging. The amount of electricity the lights use should be relatively low compared to the amount needed to charge a car. I also considered using a baseline calculated as the lowest usage during an hour of the day to avoid this problem. The results were similar between the two, and the latter option increased the probability that I set the baseline too high for station-hours that are consistently busy. It is much more likely that a station always has customers at 2 PM than that it always has customers at all hours.

The panel of station-week-hour charging contains 1.5 million observations for 468 stations over 209 weeks. 178 stations are Tesla stations, 128 are EVgo, 90 are Electrify America, and 72 belong to other networks. I am able to match all to price data, but can only include 115 Tesla, 103 EVgo, and 38 Electrify America stations in the demand estimation model. There are two reasons for the large drop in stations for Tesla and Electrify America. First, Electrify America installed batteries at many of its stations and I omit all stations with

batteries installed because batteries are charged in the late morning (when electricity is cheap) and discharged in the late afternoon (when it is expensive), which will bias estimates. Second, I limit the model to just CBSAs with at least one station from each network and at least 10 stations overall. Many Tesla stations are along highways in more rural areas and therefore are not included in the deamnd estimation.

C Data Appendix

Technical Appendix

Proof of profit maximizing price

Consider a consumer that must choose where to charge among n stations indexed by i. The consumer's preference for charging at each station is given by the vector $\boldsymbol{\theta} = (\theta_1, ..., \theta_n)$ where θ_i includes preferences over nearby amenities (such as grocery stores, restaurants, etc.), charging network, charging standard, and travel costs. The vector of prices at each station is $\boldsymbol{p} = (p_1, ..., p_n)$. The consumer chooses to charge at the station that maximizes net surplus $s_i = \theta_i - p_i$ for $i \in 1, ...n$.[38]

For simplicity, let all consumers' preferences for stations be independent and identically distributed $\theta_i \sim U[-1, 1]$. The cumulative distribution function (CDF) is $G(\theta_i) = \frac{\theta_i + 1}{2}$. The probability that a consumer prefers station i to station j is

$$\Pr(s_i \geq s_j) = \Pr(\theta_j \leq \theta_i + p_j - p_i) = G(\theta_i + p_j - p_i) = \frac{\theta_i + p_j - p_i + 1}{2}. \tag{14}$$

Aggregating over L consumers who each charge at the station where they get the greatest surplus yields station i's quantity:

$$Q_i(\boldsymbol{p}) = \Pr(s_i \geq \max_{j \neq i} s_j)L = L \int \Pi_{j \neq i}[G(\theta_i + p_j - p_i)]g(\theta_i)d\theta_i. \tag{15}$$

Let each station face constant marginal cost c. Then its profit function is

$$\pi_i(\boldsymbol{p}) = (p_i - c)Q_i(\boldsymbol{p}) - K = (p_i - c)L \int_{-1}^{1} (\frac{\theta_i + p_j - p_i + 1}{2})^{n-1} \frac{1}{2} d\theta_i - K \tag{16}$$

where K is the station's fixed cost of entering. Holding competitors' prices constant, the profit maximizing price for each station is:

$$p_i = c - \frac{Q_i(\boldsymbol{p})}{\partial Q_i / \partial p_i}. \tag{17}$$

Assuming symmetry yields the Nash-Bertrand equilibrium. All stations charge the same

[38] For simplicity, I have omitted the outside option of not charging at any station. Adding in the outside option sets an additional restriction that $s_i \geq \bar{v}$ for the consumer to choose station i.

price $p_i = p$ and sell the same quantity $Q_i = L\int_{-1}^{1}(\frac{\theta_i+1}{2})^{n-1}\frac{1}{2}d\theta_i = \frac{L}{n}$. The slope of the demand curve is $\frac{\partial Q_i}{\partial p_i} = -L\int_{-1}^{1}(n-1)(\frac{\theta_i+p-p_i+1}{2})^{n-2}\frac{1}{4}d\theta_i = -\frac{nQ_i}{2} = -\frac{L}{2}$. Plugging into p_i yields the equilibrium profit maximizing price

$$p(n) = c + \frac{L/n}{L/2} = c + \frac{2}{n}. \tag{18}$$